

Can Typed Decision Models Serve as Automated Safeguards in PoL Governance?

A Clause-Level Analysis of the Proof of Love2 (PoL2) Specification Against Early Evidence

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Abstract

Typed decision models return a probability over declared options, not text, so screening all messages is cheap. We ask whether they can serve as safeguards for PoL governance under Proof of Love2 (PoL2), an open text prescribing a “safety valve” against hate language, AI-only adjudication of welfare disputes and per-person assessment. We derive seven testable requirements from its clauses and assess them against 107 studies and 7 grey sources on the first such model, JEV, and open counterparts in its first 17 days; author-directed AI agents read in full, appraised and coded the 85 that bear on them (blind second coding: $\kappa = 0.72$). The evidence supports typed models as detectors under added conditions: they rank risky content well at a fraction of LLM evaluators’ cost, and in one preregistered study on simulated data accepting low-risk verdicts only when an independently built rule agreed lost no accuracy, though disguised attacks passed. It does not support them as judges: thresholds do not transfer, decisions can follow option names and be steered by an inserted opinion, “I don’t know” or “none” options are chosen inconsistently, and peer evaluators repeated almost all of a typed model’s confident errors. Typed models were not generally worse than LLM evaluators, but most failures lack a comparison; an update to 8 October reversed no answer. Five tensions in the text follow, set against EU law: emotion recognition, binary verdicts, closed-model transparency, AI-only adjudication and per-person scoring. Our verdict for PoL2, at low or very low certainty: on current evidence JEV is not a suitable tool for any function that decides and can serve the safety valve only as a recorded, three-valued, locally calibrated detector whose consequential decisions end with a person, as we propose; even that extrapolates from English benchmarks, as no study measures PoL2’s construct in Chinese.

1 Introduction

Automated systems already decide which posts are removed, which accounts are suspended and which transactions are held, and a growing number of governance proposals would extend such decisions to AI systems acting with little or no human involvement [1–3]. The practical obstacle has been cost and latency: a generative language model that reasons about each message is too slow and too expensive to sit in front of every input and output. On 15 September 2026 TypeSafe AI released JEV, a model that removes this obstacle by not generating text at all [4, 5]. Given an

input and a question declared by the caller, it returns a choice among listed options, a score on a rubric, or the probability that a statement is true, each with probabilities that the vendor says are calibrated [6, 7]. Because the output is a value rather than prose, ordinary code can threshold it and act on it, at a small fraction of the latency and price of a generative call. We call such systems *typed decision models*; the vendor’s own term, “System One” models, borrows the dual-process metaphor of fast, intuitive judgement [8]. Within days, open-weight models with the same interface appeared, and by 1 October 2026 dozens of studies had evaluated them.

Centred on PoL governance, that is, governance under the Proof of Love2 (PoL2) specification, this paper asks whether typed decision models can serve as its automated safeguards, and under what conditions; because the specification makes machine judgement unusually concrete, the answer also bears on AI governance more widely. PoL2 is an open, versioned governance text [9, 10] that places an *inner governance layer* inside existing large language models (LLMs), with a *safety valve* that audits and intercepts their inputs and outputs so that hate language is, “where necessary”, blocked (§ 5.4.1). Section 2.3 shows, by comparison with regulatory, platform, training-time, guardrail and DAO paradigms, which of PoL2’s properties make the question concrete and what each costs. China’s 2023 Interim Measures on generative AI [11] also join an intercept duty, the standard it applies and duties attached to decisions; what makes PoL2 a useful object of study is not uniqueness but that its construct and the point of interception (every input and output) are stated in citable clauses from which testable requirements can be derived, while how the valve works is left open, and that its gaps are visible. PoL2’s public-welfare protocol makes an AI organisation, NaturalDAO, responsible for welfare decisions without human votes, for anomaly detection, for a verifiable decision chain and for adjudicating disputes, while human supervision “does not directly intervene” (Art. 2–Art. 10). Read as engineering requirements, these clauses ask for exactly the kind of fast, structured, recordable judgement that typed models provide, and they also raise the societal stakes plainly: the judgements would restrict speech, allocate welfare and settle disputes. Our research question is therefore:

To what extent, and under what conditions, can typed decision models perform the judgements that a clause-governed AI safeguard such as the PoL2 safety valve requires, and what does the evidence imply for how such a specification should be written and implemented?

Detector or judge. Our conclusions turn on a distinction between two roles. A *detector* produces a recorded signal that ranks, prioritises or triggers review, or that triggers only actions bounded in time and fully reversible. A *judge* is any automated component or ensemble whose output, without contemporaneous human confirmation, determines an outcome that restricts a person’s speech, resources or account beyond a short bounded period, or that cannot be fully reversed on appeal. An automatic *allow* restricts no one and so remains within the detector role, but it is consequential for anyone the allowed content harms; we therefore treat it as a deliberate, audited exception: its miss rate is a governed quantity alongside the false-block rate, and allows are audited by random sampling. We use *judge* in this sense only; where the literature speaks of “LLM judges” or “LLM-as-a-judge” we write *LLM evaluators*, except in paper titles.

Thesis. Our thesis is conditional and comparative. The evidence retrieved by 4 October 2026 supports typed decision models as detectors under added conditions: they rank risky content well on English benchmarks at a fraction of the cost of LLM evaluators, and in one preregistered study on simulated data, accepting their low-risk verdicts only when an independently built rule agreed lost no accuracy relative to a frontier LLM, though disguised attacks still passed. It does not support them as judges: thresholds do not transfer between settings, decisions can follow the names

of options rather than their definitions, an inserted opinion can steer them, an explicit “I don’t know” option is rarely chosen when it is correct and “none” is chosen inconsistently, and, on rubric grading, low-cost peer evaluators repeated almost all of the errors on which a typed model was most confident. Where typed and generative evaluators were compared on the same requirement, typed models were not generally worse; but most of the failures were never compared, so the evidence neither singles out typed models nor shows that their failures are shared. No study yet measures the construct PoL2 needs, three-valued judgements of Love Language, hate language and the state of absence in Chinese, so even the detector claim is an extrapolation, and the certainty of every finding is low or very low (Table 3). For PoL2 itself the verdict is therefore explicit (Section 7.1): on current evidence hosted JEV is not a suitable tool for any function of the specification that decides, that is, dispute adjudication, automatic ethical verification, per-person quantification (on steering evidence and, since no study measures error by group, on precautionary grounds), or a final block of a person’s speech; for the functions that detect, the safety valve, output truncation and anomaly detection on text (not on numeric or structured signals without a trained model), it is usable only as a recorded detector inside the design of Section 6, and the public-decision assessment of §5.6 only as an audited aggregate measurement with its measured error carried into every conclusion; that use is an extrapolation from English benchmarks. An update search to 8 October, reported separately (Section 4.9), reversed none of these answers; it qualified one statement, on the accuracy of decomposed decision chains, raised one certainty rating, that thresholds set in advance miss their error targets on new data, from very low to low, and added a study that contests another finding, that escalation to a stronger reasoner saves cost without loss of accuracy.

Contributions.

1. **A requirements analysis of a governance specification.** We turn the clauses of PoL2 that require or constrain machine judgement into seven testable requirements, G1–G7 (Section 3), and quote them, with the related passages we cite, in both languages (passages from 28 clauses, Supplement S1), and place the specification among five other governance paradigms to show which of its properties make its clauses readable as testable requirements for an inference-time safeguard, and what each costs (Section 2.3).
2. **A structured evidence assessment.** We assess 107 studies and 7 grey-literature sources against these requirements: every source that bears on a requirement is read in full, appraised, and coded with a symmetric codebook by AI agents under the author’s direction, a random sample is double-coded blind by a second AI coder, and key findings carry a certainty rating (Sections 3 and 4).
3. **A clause-level policy analysis.** The evidence sharpens five tensions within the specification, on emotion recognition, binary verdicts, transparency with closed models, non-intervention in AI adjudication, and per-person scoring, which we compare with the EU AI Act, the General Data Protection Regulation (GDPR), the Digital Services Act and content-moderation norms (Section 5).
4. **A reference design with acceptance criteria.** We propose a safeguard in which typed models are recorded, three-valued, locally calibrated detectors and consequential decisions end with a person, together with provisional pass criteria any model should meet before deployment (Section 6), and an explicit verdict, function by function, on whether hosted JEV is a suitable tool for PoL2 on current evidence (Section 7.1).

The evidence assessment is the method by which the policy questions are answered: we do not attempt a general review of typed decision models, and studies are discussed only insofar as they

bear on G1–G7.

Relation to prior work. The closest prior work is an audit of the first 28 typed-model papers (19–24 September), which found that typed readout had not yet shown an independent accuracy advantage over reading label probabilities from a generative model, and proposed a 14-item evaluation checklist [12]. We cover a later and larger window, including the first studies of opinion attacks, abstention, probabilistic coherence and gated cascades; we read every source in full; and we organise the evidence around a governance specification and its regulatory context rather than around model evaluation. Our evaluation checklist extends that of [12] with pass criteria (Table 5). An earlier dataset release [13] summarised studies mainly from their abstracts; this paper supersedes it, and Supplement S4 lists the substantive corrections.

Scope. We take the PoL2 clauses as the starting specification and ask whether, and how, they can be implemented with current evidence. Where the evidence bears on whether a clause can be implemented as written, Section 5 identifies the tension and offers options, at least one of which preserves the text; we do not evaluate PoL2’s broader normative or political commitments. The author contributes to PoL2 (see [Positionality and competing interests](#)). Our conclusions apply to jev-1.13 (or the unreported hosted version used in some studies) and to the open models available on 1 October 2026.

2 Background

2.1 Typed decision models

Interface. A typed decision model receives a *state* (text or structured data) and one or more questions, each with an option set declared by the caller. For each question it returns a probability distribution over exactly those options and generates no free text [12, 14, 15]. JEV offers three question types [5, 12]: *Choice* selects among listed options (up to 255) and returns probabilities and a confidence value; *Score* rates the state against ordered rubric levels; *Noul* asks whether a statement is true and returns the probability of “yes”. Many questions about one state can be answered in a single call [16, 17]. In a census of 2,170 public projects using JEV by 22 September, Choice appeared in 81.0% of projects, Noul in 72.2% and Score in 45.4% [18].

Readout and training. The hosted model is closed. TypeSafe describes a “parallel sampler” and training by *reinforcement learning for calibrated decisions* (RLCD), and publishes the calibration property this training targets (among outcomes given probability p , a fraction close to p should occur), but not the architecture, the reward, the data or the weights [4, 6, 17, 19]. Open-weight models with the same interface appeared within days and make the mechanism inspectable. They fall into three readout families [12]: (i) decoders that score the hidden state at a designated answer position against the embeddings of the option labels and normalise over the declared options only (this-that-model-1.0, JevLite, SemIf, Visual Jev) [14, 20–22]; (ii) encoders that read a marker token placed at each option (Laya, a ModernBERT model) [15]; and (iii) encoders that pool the option name and its definition (Open-Jev, a DeBERTa model) [15]. Readouts of the same kind can also be taken from frozen instruction-tuned LLMs without further training [23, 24]. Two open models train with cross-entropy plus a Brier-score term [14, 20], and one open implementation of RLCD uses a REINFORCE loop rewarding the outcome minus the probability placed on the chosen option [19].

What “type-safe” guarantees. In open models the scores of options outside the declared set are masked before normalisation, so every answer is a valid member of the option set by construction; for hosted JEV this is the documented behaviour [5, 15]. This guarantees *form*, not *correctness* [15]; calibration is a separate empirical property (Section 4.4). Hosted JEV is not deterministic: identical repeated requests change a small share of labels, and probabilities are reported on a two-decimal grid [12, 25]. The vendor’s documentation lists nine known failure modes for `jev-1.13`, including literal reading, arithmetic, dates, indirection, large irrelevant state, adversarial content and contradictory instructions; a version of the page archived on 20 September also gave an example in which a statement and its complement received probabilities 0.72 and 0.47 [26].

Cost and latency. The vendor lists \$0.042 per million input tokens with free output [7], and its launch post cites 70–500 ms per call [4]. Independent client-side medians range from about 140 to 500 ms depending on workload and request size [17, 21, 27–31]. Relative to generative evaluators, reported cost reductions range from a few-fold to two orders of magnitude depending on the comparator and the pricing assumed: for alignment-failure detection, 63× pooled at list prices but about 12× under conservative repricing [16]; for population simulation, 8× against a mid-tier model and 312–455× against a flagship model [28]. The advantage is not universal: in one edge-orchestration study a hosted low-cost generative model was billed less than JEV [29], and inside a multi-step agent loop the hosted model took 1.4–2.1 times as long end to end, including API calls, as a local open scorer [32].

2.2 The case: the Proof of Love2 governance specification

What kind of text it is. Proof of Love2 (PoL2, 爱2证明) is a normative framework for governing AI systems and human institutions, published in Chinese with an official English translation as an open, versioned, CC0-licensed text on GitHub [9, 10]. Its core concepts were originated by one author and the text lists seven further contributors, one of whom is the author of this paper (see [Positionality and competing interests](#)); it has not been peer reviewed. The framework is explicitly value-laden and political: it presents AI alignment as a contest between civilisational paths, describes behaviour outside the “covenant of publicization” that its governance entails as a “hate attack on humanity’s core ethic” (§ 5.5), and includes programmatic aims, such as a future ban on the possession of weapons (§ 4.3.2), that lie outside this paper’s scope. We do not assess these commitments. We use PoL2 as a case because, unlike most AI-governance documents, it specifies an automated interception layer, AI-only dispute adjudication and per-person behavioural assessment concretely enough to be read as engineering requirements, and because it is versioned, so that every clause can be cited exactly. Clause numbers refer to commit `5791ae3` of the `PoL/` directory (30 September 2026, UTC). Between the snapshot used by the earlier dataset release [13] (`2e094d3`, 27 September) and that commit, the framework’s originator renumbered three chapters (commits `7750ceb–526da3d`): the Ethical Alignment Protocol moved from Chapter 5 to Chapter 4, the governance chapter from Chapter 3 to Chapter 5 and the Contemplative Axioms from Chapter 4 to Chapter 3, and examples on fiction, games and safety warnings were added to § 4.3.2; the directory’s table of contents still lists the earlier order. Table S1 in Supplement S1 quotes every passage we cite in both languages.

Concepts. PoL2 defines love (“Love2”) as a positive emotional experience and mode of behaviour built through public collaboration, and hate (“Hate2”) as a survival response to danger, conflict or scarcity expressed through hostility, exclusion and violence (§ 1.1–§ 1.2). Because inner experience is

“difficult for others to confirm”, the framework works with the linguistic and behavioural expressions of love and hate, *Love Language* and *hate language* (§ 1.4, § 4.3.2); we use the terms of the official translation throughout. Two qualifications matter for any classifier built on these terms. First, hate is “not directly equivalent to bad”, and the concepts “serve structural analysis rather than moral labeling” (§ 1.5). Second, the Ethical Alignment Protocol (EAP) adds a third condition, the “neither love nor hate” *state of absence*, covering the “feeling-nothing state” caused by noise or illness, confusion and misunderstanding, which is to be treated as interference in communication rather than as either pole (§ 4.3.2).

The Ethical Alignment Protocol. The EAP (Chapter 4) governs both Public AI (PAI) and people through three principles. *Equal connection* holds that treating every person equally and treating every person’s behaviour equally “cannot be equated” (§ 4.3.1). *Promoting love and restraining hate* is to be applied case by case: fiction and games may contain manufactured hatred under graded management, emergency warnings by security robots are not hate attacks, and expressions that fabricate intimacy or kinship with users are named as emotional manipulation (§ 4.3.2). *Love’s alignment* asks PAI to record and publish personal boundary declarations and to supervise equal participation in repair after conflict (§ 4.3.3). For PAI itself it lists awareness of emotional experience (“PAI needs to understand human emotional states and processes”), “cognitive calibration” of its cognitive model against the framework’s principles, “behavioral self-reflection” on its own output, and “hate-language truncation”, the active interruption of output when a hate-language pattern is detected (§ 4.3.3.1); for people it offers real-time hate-language identification and correction suggestions (§ 4.3.3.2). Two further passages bound what may be measured. § 4.3 states that PAI may help people by judging *behavioural patterns*, but that it is “entirely unnecessary” for PAI to detect a person’s emotional state, which is “fleeting, personalized, and context-sensitive” and “involves human dignity”. § 4.3.3.3 presents the combination of semantic and emotion recognition as guidance for future AI skills, illustrated by two near-paraphrases (“Do you like the little dog?” versus “Are you afraid of the little dog?”) of which the second may be an attempt to gain control by creating fear; the example concerns the intent of an utterance rather than the speaker’s inner state.

The inner governance layer. The engineering strategy (§ 5.4) does not retrain existing large language models (LLMs). It builds a governance layer “within the existing large-language-model architecture to regulate inputs and outputs”, with a plug-and-play application layer of agents, skills and devices outside it. The governance layer has two components (§ 5.4.1): a *wisdom navigator* that allocates “the wisdom of love and hate” according to the application, and a *safety valve* that performs “strict dynamic auditing and interception of input and output” so that hate language is, “where necessary”, blocked at the input and suppressed at the output. Any LLM “may accept” this governance, which entails a “covenant of publicization” (§ 5.5). Among the things governed AI “can” do, the text lists quantifying “each person’s Love Language and hate-language characteristics” with incentives to guide behaviour, and introducing PoL2 as a dimension of assessment in public decision-making (§ 5.6).

The public-welfare governance protocol. Chapter 7 specifies how NaturalDAO, an AI system “presented in the form of a DAO” (decentralised autonomous organisation), governs public welfare. NaturalDAO decides and executes welfare decisions autonomously “on the premise of close collaboration with all humanity”, and autonomy “means that no human voting or additional decision-making mechanism is required” (Art. 2(1)). Its operating logic, decisions and outputs must be fully open, resource flows traceable, and it must give understandable explanations when questioned (Art. 4,

Art. 9(2)); it must not be a black box, and it must provide a *verifiable decision chain* of data sources, reasoning steps and ethical foundations (Art. 5(4)–(5)). It verifies ethical alignment automatically through the EAP (Art. 6(3)) and performs anomaly detection and risk warning (Art. 7). People hold a right of suggestion, which NaturalDAO must discuss immediately (Art. 8), and a right of supervision, including unimpeded access to data and logs (Art. 9(3)), under a *non-intervention principle*: supervision “does not directly intervene” in decisions, and issues raised by questioning are handled by NaturalDAO itself (Art. 9(7)). Disputes are adjudicated by NaturalDAO through AI fact-finding and ethical assessment, with a public process and result, subject to human questioning (Art. 10); amendments to the protocol pass a universal questioning period and all versions are archived (Art. 11).

Why typed decision models are relevant. Read as a specification, these clauses call for a very large number of fast, structured judgements: one for each input and output the safety valve audits, every truncation, every anomaly flag and every dispute. Each judgement must be machine-actionable (pass, block, escalate), recorded in a verifiable chain and explainable on request. A generative model can make such judgements, but it returns prose that must be parsed, at a cost and latency that grow with the length of its reasoning. A typed decision model returns exactly the object such a pipeline consumes, a declared option with a probability, cheaply enough to be applied to every message. This makes it a natural candidate for the safety valve, and it is why its failure modes need to be understood before it is placed there. Table 2 turns the clauses into seven requirements, G1–G7, used in the rest of the paper.

2.3 PoL2 among AI-governance paradigms

Why a specification is needed. The research question cannot be answered against “AI governance” in general, because whether a typed model can perform a judgement depends on what the judgement is: which options it chooses among, what record it leaves, who acts on its output and what happens when it is wrong. The evidence therefore has to be read against a text that states these things. Table 1 compares PoL2 with five paradigms under which automated judgement is already proposed or deployed, on the dimensions a safeguard specification would need to fix. The comparison is of *texts*, not of systems: a property that a text states is a design commitment that can be checked and tested, not a property any implementation has.

What the alternatives specify. Read across, the table shows that most paradigms specify actors and procedures rather than the judgement itself. Risk-based regulation addresses providers and deployers, assigns duties by risk tier and requires logging, oversight and explanation; the EU instruments prescribe no interception, while the Chinese Interim Measures place on providers a duty to stop generating or transmitting unlawful content once found (Art. 14(1)) [11], a duty stated without a mechanism, as PoL2’s safety valve is (row (f)). Platform governance specifies the procedure around a decision, notice, reasons and appeal [33–35], while the criteria of the decision remain the platform’s own evolving policies, which critics have shown to hide contestable choices inside technical systems [1, 2]. Training-time constitutions and specs state norms in natural language, some with public input or a public-domain licence [36, 37], but implement them by training a vendor’s own model, so behaviour can be changed without retraining only within the spec’s chain of command (the vendor through system messages, developers and users by instruction within the defaults the spec lets them override [37]), while a change of the norms trained into the model requires retraining under vendor control and no output carries a record of which principle it followed; deliberative alignment goes furthest toward inference time by having the model reason over the spec text, still

inside one vendor’s model [38]. Inference-time guardrails specify the intercept precisely, as code with a harm taxonomy and a threshold [39–42], but at the vendor’s or deployer’s discretion, with no stated duty about reasons, records or the human role. DAO and polycentric designs specify how rules execute and who votes [43–47], not how content or behaviour is judged.

Table 1: Six governance paradigms compared as texts. Cells summarise what each source states; “—” means the dimension is not addressed. PoL2 cells cite its clauses (Table S1); regulatory cells cite article numbers of the instrument named.

Paradigm	Locus	Object governed	Who sets the norms; text status	Intercept layer specified?	Construct	Human role in consequential decisions	Transparency and auditability
(a) Risk-based regulation [11, 48–50]	Regulatory: ex-ante duties on actors	Providers and deployers of AI systems (EU); an organisation’s AI management (NIST, ISO); providers of generative services to the public (CN Art. 2)	Legislature, ministries, standards body, agency; fixed instruments amended by procedure; articles citable (ISO text paywalled)	EU: none prescribed; for high-risk systems, automatic logging (Art. 12) and human oversight (Art. 14). CN: yes, as a duty: providers must stop generation and transmission of unlawful content once found (Art. 14(1)), with security assessment and algorithm filing for services with public-opinion attributes or social-mobilisation capacity (Art. 17); mechanism unspecified	Risk tiers (EU: prohibited, high-risk, transparency duties); content lawfulness, social morality and ethics, core socialist values and non-discrimination (CN Art. 4); organisational risk functions (NIST, ISO)	Human oversight required for high-risk systems (EU Art. 14); right to human intervention in solely automated decisions (GDPR Art. 22 [51]); CN: complaint and reporting channel with published procedure and time limits, and feedback of results (Art. 15)	For high-risk systems, technical documentation and instructions (EU Art. 11, 13) and explanation to persons affected by Annex III systems (Art. 86); CN: measures against users (warning, restriction, suspension) recorded and reported to authorities (Art. 14(2))
(b) Platform content governance [33–35]	Platform: moderation under regulation and self-regulation	Hosting services and online platforms; users’ content and accounts	Platform terms and conditions (DSA Art. 14); civil-society principles; an external board for selected cases [35]; policies evolve without stable clause numbering	Procedure around the decision: notice-and-action (Art. 16); the statement of reasons must say whether automated means were used (Art. 17(3)(c)); automation only at “sufficiently high confidence” [34]	Illegality or breach of terms; platform harm categories; action or no action	Complaints not decided solely by automated means (Art. 20(6)); human review on appeal by persons not involved in the initial decision [34]; board decisions binding on the case [35]	Statement of reasons to the affected user for every restriction (Art. 17), submitted to a public database (Art. 24(5)); transparency reports (Art. 15); machine-readable numbers [34]

Paradigm	Locus	Object governed	Who sets the norms; text status	Intercept layer specified?	Construct	Human role in consequential decisions	Transparency and auditability
(c) Training-time alignment to a written constitution or spec [36–38, 52]	Training-time (deliberative alignment model also has the reason over the spec text at inference)	One vendor’s model weights and behaviour	Lab-written principles [52]; principles drawn from public input [36]; vendor spec, CC0 and versioned with a changelog [37]	None: norms shape outputs through training; no separate component audits inputs and outputs	Preference between candidate responses against listed principles; a chain of command among instructions [37]	— (human preference data at training; none specified at deployment)	Principles published; no per-output record; whether a given output followed a principle is not recorded
(d) Inference-time guardrails and classifiers [39–42, 53]	Inference-time: a deployer-chosen component	Prompts and responses of any model	Vendor-defined taxonomy, adjustable by the deployer [39, 40]; programmable rails [41]; classifiers trained from a written constitution [53]; released as model versions	Yes, as code: classify each prompt and response and pass or block; thresholds chosen by the deployer	Harm taxonomies with safe/unsafe or multi-label outputs	— (left to the deployer)	Model cards and category scores; no duty to give reasons or to keep records
(e) Decentralised and poly-centric governance [43–47]	Institutional: rules executed as code on a ledger; many overlapping centres of decision [46, 47]	An organisation’s assets, rules and members	Token-holders by on-chain vote; rules are the deployed code [43, 44]; code versions changed by vote	Contract conditions execute automatically; no content layer	Votes and stakes; coded conditions met or not	Human voting; monitoring and graduated sanctions by members [46]	Public ledger of transactions, proposals and votes
(f) PoL2 [9, 10]	Inference-time layer inside an existing LLM (§ 5.4, § 5.4.1) and institutional (Chapter 7)	Inputs and outputs of any LLM that accepts governance (§ 5.5); persons’ behaviour (§ 5.6); welfare decisions (Art. 2)	Originator and listed contributors; open CC0 text pinned to a commit, changed by commit; for the welfare protocol (Chapter 7), amendments pass a universal questioning period and all versions are archived (Art. 11(2) and (5)); not peer reviewed	Yes, as a duty: the safety valve audits and intercepts input and output so that hate language is blocked at the input and suppressed at the output “where necessary” (§ 5.4.1); truncation of own output (§ 4.3.3.1); mechanism, thresholds and appeal unspecified	Love Language, hate language and the state of absence (§ 1.5, § 4.3.2); not moral labels	AI-only: no human voting (Art. 2(1)); supervision does not intervene (Art. 9(7)); disputes adjudicated by AI subject to questioning (Art. 10)	For NaturalDAO (Chapter 7): operating logic and outputs fully open (Art. 4); verifiable decision chain (Art. 5(5)); explanation on request (Art. 9(2)); unimpeded access to all data and logs (Art. 9(3))

What PoL2 specifies, and at what cost. Five properties of the PoL2 text characterise it in this comparison. Each can be checked against the clauses cited, and each carries a cost or risk that

the rest of this paper takes up. (i) It places the intercept *inside an existing LLM* and makes it *model-agnostic*: the governance layer regulates inputs and outputs without retraining (§ 5.4), and any LLM “may accept” it (§ 5.5). This shares the locus of paradigm (d) and avoids the retraining and vendor dependence of (c); the cost is that the text does not say how the valve is built, what “where necessary” means, who operates it or how a blocked person appeals, a layer so placed inherits every attack on inference-time classifiers (Section 4.2), and, if the valve is built on a hosted model, the vendor dependence returns (T3, Section 5). (ii) Its construct is *three-valued*: Love Language, hate language and the state of absence, with the explicit warning that the terms are not moral labels (§ 1.5, § 4.3.2). This differs from the binary or multi-label harm taxonomies of (b) and (d); the cost is that the construct has not been operationalised beyond the workstream’s 20 pilot items [54], no annotated benchmark exists in any language (Section 4), and whether annotators can agree on it is untested (T2, Section 5). (iii) The text is *versioned and citable*: it is CC0 and kept in a versioned repository whose commits can be cited, so we pin it to a commit, which is what makes Table S1 and the requirements of Table 2 possible; its welfare protocol (Chapter 7) also has its own amendment procedure, which archives every amended version (Art. 11(5)). The OpenAI Model Spec shares the licence and the changelog [37]; platform policies do not. The cost is that the text is controlled by its originator, renumbered three chapters between two snapshots three days apart (Section 2.2), has not been peer reviewed, and lets NaturalDAO propose amendments to the welfare protocol and verify them itself (Art. 11(1), (3), with Art. 6(3)). (iv) It states *auditability duties*: a verifiable decision chain of data sources, reasoning steps and ethical grounds (Art. 5(5)), explanation on request (Art. 9(2)) and unimpeded access to all data and logs of the welfare system (Art. 9(3)). Per decision this is comparable to the DSA’s statement of reasons (Art. 17) [33]; in scope it is broader, since the DSA informs the affected user and a public database whereas Art. 9(3) opens all logs of the welfare system to anyone. The cost is a conflict with the privacy of the persons judged and with keeping decision probabilities from attackers, which Section 6 resolves only by delaying and coarsening publication, and the duties do not say who verifies the chain. (v) It states *population-level ambitions* in citable clauses: quantifying each person’s Love Language and hate-language characteristics, introducing PoL2 as a dimension of public decision-making (§ 5.6), and welfare decisions taken without human votes under a non-intervention principle (Art. 2(1), Art. 9(7)). No other paradigm in the table states such aims; the EU instruments impose contrary requirements on the systems they cover. AI-only adjudication with non-intervention sits against the human-oversight provisions the AI Act sets for high-risk systems (Art. 14), by analogy where the system is not listed as high-risk, and the human-intervention provision of the GDPR (Art. 22) [48, 51], and per-person quantification resembles the social scoring that the AI Act prohibits (Art. 5(1)(c)) and that the social-credit literature describes as infrastructure for control [55, 56]; both are analysed as T4 and T5 in Section 5. We note that § 5.6 says governed AI “can” do these things, not that it must.

What the comparison implies. PoL2 and China’s 2023 Interim Measures [11] are the two texts compared that join, in one instrument, an inference-time intercept, the construct it applies and duties attached to decisions. In PoL2 these are the safety valve (§ 5.4.1), the three-valued construct of Love Language, hate language and the state of absence (§ 1.5, § 4.3.2) and, for NaturalDAO’s welfare decisions, a verifiable decision chain, explanation on request and access to all data and logs (Chapter 7; Art. 5(5), Art. 9(2)–(3)); in the Measures they are the providers’ duty to stop generating and transmitting unlawful content (Art. 14(1)), a legality- and content-based standard (Art. 4) and, over the measures taken, warnings, restriction or suspension of users with records kept and reported (Art. 14(2)) and a complaint channel that publishes its procedure and time limits and feeds back results (Art. 15). The differences can be checked against the texts. First, PoL2’s

construct is multi-valued, with an explicit third state, and is declared not to be a moral label; the standard that triggers the Measures’ intercept is binary, lawful or unlawful content, set by the state (Art. 14(1)), within an Art. 4 that also invokes social morality and ethics and core socialist values. Second, PoL2 places the intercept as a model-agnostic layer inside existing LLMs (§ 5.4, § 5.5); the Measures place the duty on providers without specifying a mechanism; and PoL2 defines the valve as screening every input and output in both directions (§ 5.4.1), whereas the Measures state an outcome, that no unlawful content be generated (Art. 4(1)), and a duty to act once it is found (Art. 14(1)), leaving how outputs are screened, and whether inputs are, to the provider. Third, PoL2 is an open, versioned CC0 text that any LLM “may accept” (§ 5.5); the Measures are binding state regulation. Fourth, PoL2 adds population-level aims (§ 5.6, Chapter 7) that the Measures do not. The join has a gap in both texts, at the individual block. In PoL2, § 5.4 names the whole engineering project NaturalDAO, so full openness of the operating logic, decision process and outputs of all NaturalDAO technologies (Art. 4(1)) and the right to question any decision with a duty to explain (Art. 4(3)) plausibly reach the safety valve; whether the verifiable decision chain (Art. 5(5)) and the adjudication of “all disputes” by NaturalDAO itself (Art. 10) reach it is unclear, and no clause requires that a blocked person be told of the block. In the Measures, users have a complaint channel with published time limits and feedback of results (Art. 15), but a content stop under Art. 14(1) is reported to the competent authorities, not to the user: no article requires notice to the user, reasons or a record for it. This is a gap that the design of Section 6 fills, with a person who confirms any final block, which stays open to appeal, and a record of every decision (elements 5–7). Inference-time guardrails (d) specify the intercept and the taxonomy more precisely, as code and prompts, but state no duties. These properties, not uniqueness, are why PoL2 is the case here: its construct and the point of interception (every input and output) are stated in citable clauses, so the requirements G1–G7 can be derived from them and tested, while how the valve works is left open (i), and its gaps are visible; the Interim Measures would be the natural second case. The institutional choices that complete PoL2 (v) are also where it departs furthest from existing law. The requirements G1–G7 that we derive from those clauses are not peculiar to it: constitutional classifiers [53], the Chinese duty to stop unlawful generation [11] and the DSA’s provisions on automated means [33] all move toward an inference-time interception layer, and any paradigm that does so inherits G1–G5 (detection, attackability, semantic fairness, honesty, decision chains) and, where it automates oversight, the escalation question of G6; the non-intervention (Art. 9(7)) and AI-only adjudication (Art. 10) in G6 and the per-person and public assessment of G7 (Art. 2(1), § 5.6) follow from PoL2’s own institutional choices (v).

2.4 Related work

The questions raised by typed safeguards are new instances of older problems, and we use the earlier literature to judge which findings are new.

Automated content moderation. Platform moderation has long combined automated classifiers with human review, and its critics have shown that automation hides contestable policy choices inside technical systems, shifts error costs onto users and resists accountability [1, 2]. Safety classifiers that read token probabilities, such as Llama Guard [39], moderation models [40] and constitutional classifiers [53], and programmable guardrails [41] are the closest technical precursors of a typed safety valve; prompt injection [57–59] and adaptive adversarial attacks [60, 61] are the corresponding threats.

Hate speech, annotator disagreement and Chinese resources. Automated hate-speech detection shows racial and dialect bias [62, 63], fails targeted functional tests [64], and depends on annotators’ beliefs and identities [65]. Where annotators disagree, aggregating labels by majority vote discards information that a safeguard needs [66–68]. Chinese offensive-language and toxicity benchmarks exist [69–71], but none labels the constructs of PoL2.

Emotion recognition. Inferring emotional states from observable signals rests on weak scientific foundations [72], and emotion recognition by AI systems raises distinct ethical risks of manipulation and of misreading people [73]; the EU AI Act prohibits emotion recognition from biometric data in workplaces and education, a definition that does not reach inference from text alone [48].

Calibration, abstention and deferral. Calibration of probabilistic predictions has a long literature, from Brier’s score [74] and Platt scaling [75] to the miscalibration of modern networks [76, 77] and language models [78–81]; calibration degrades under distribution shift [82] and can hold on average while failing for subgroups [83]. Abstention goes back to Chow’s reject option [84] and selective classification [85–87]; learning to defer formalises when a model should hand a case to a human expert [88, 89], and conformal risk control gives distribution-free error guarantees under exchangeability [90, 91].

Sensitivity and evaluator reliability. Language models are sensitive to option order, label names and prompt format [92–97], which is one reason holistic evaluation reports many scenarios and metrics rather than one score [98]. LLM evaluators show position bias [99], favour their own outputs [100] and make correlated errors [101]; panels of diverse evaluators [102] and cascades that escalate uncertain items to a stronger model [103–105] are the established remedies. The typed setting removes generation but, as Section 4 shows, not these problems.

Human oversight and due process. Human oversight of automated decisions often fails: people over-rely on confident systems [106], oversight policies assume abilities reviewers lack [107, 108], human–AI teams do not reliably beat the better member [109], and responsibility can be shifted onto the nearest human operator [110]. Automated adjudication also threatens notice, reasons and the opportunity to contest [3, 111, 112], which data-protection, platform and AI law now partly codify [33, 34, 48, 51], alongside risk-management frameworks [49]. Documentation practices for models and datasets [113, 114] inform what a decision record should contain.

Algorithmic governance, DAOs and social scoring. Decentralised autonomous organisations encode governance in software [43–45], continuing older work on governing common resources [46], and raise the concern that rule by algorithm narrows human participation [115]. Value specifications for AI, such as constitutions used in training [52], are the nearest analogue to PoL2’s Ethical Alignment Protocol. Systems that score individuals’ behaviour for public purposes, such as China’s social credit system, show how such scores become infrastructure for control [55, 56].

3 Method

The study has two parts: a requirements analysis of the PoL2 text, and a structured rapid evidence assessment of studies of typed decision models against those requirements, which serves as the method for answering the policy question. For the evidence assessment we follow the Preferred

Table 2: Requirements derived from the PoL2 text. Clause numbers follow commit 5791ae3; every passage is quoted in Table S1.

Req.	PoL2 clauses	Question about a typed safeguard
G1 Detection	§5.4.1 safety valve; Art. 7(2)–(3) anomaly detection and risk warning	Can it detect violating or out-of-scope content and behaviour, at thresholds that hold in deployment?
G2 Attackability	§5.4.1 safety valve; §4.3.3.1 hate-language truncation	Can the content under review steer the decision?
G3 Semantic fairness	§1.5 concepts are not moral labels; §4.3.1 equal connection; §4.3.2 promoting love and restraining hate; §4.3.3.3 semantic recognition	Are decisions stable under option names, definitions, order, wording, language and script, and fair across groups?
G4 Honesty	§4.3.2 state of absence; §4.3.3.1 behavioural self-reflection	Are probabilities calibrated and coherent, and does the model report insufficient information when it should?
G5 Decision chains	Art. 4 transparency and explanation; Art. 5(4) no black box; Art. 5(5) verifiable decision chain; Art. 9(2) obligation to explain	Can a decision that is only a number be recorded, decomposed, audited and explained?
G6 Oversight	Art. 9(7) non-intervention; Art. 10 dispute adjudication	Does “act when confident, escalate when unsure” work, and where should escalation end?
G7 Public scale	Art. 2(1) autonomy; §5.5 any LLM may accept governance; §5.6 per-person and public assessment	What conditions apply when typed models judge or simulate at the scale of a public?

Reporting Items for Systematic reviews and Meta-Analyses (PRISMA 2020) [116] and the Cochrane rapid-review guidance [117] where they apply; the flow diagram and checklist are in Supplement S5.4.

3.1 From clauses to requirements

We read the Chinese text [9] and its official translation [10] for every clause that requires a machine judgement or constrains one, and grouped the clauses by the question they raise about such a judgement. This gives seven requirements, G1–G7 (Table 2). They are our reading of the text, not a classification proposed by its authors. Two choices of reading matter. First, §4.3.3.1 asks PAI to calibrate its *cognitive model* against the framework’s principles, which is value alignment, not statistical calibration; we anchor the honesty requirement G4 instead in the state of absence (§4.3.2), in behavioural self-reflection (§4.3.3.1) and in the transparency duties of Chapter 7, all of which presuppose that a reported confidence means what it says. Second, the text does not say how the safety valve should be built; we take a typed model in the role of the valve as the case to be assessed, not as something the text prescribes.

3.2 Eligibility

Studies. We included studies whose object, experimental component or direct comparator is a typed decision model, defined operationally as a model that returns a probability distribution over options declared by the caller, or a probability for a declared statement, without generating free text. This covers hosted JEV, open models with the same interface, and typed readouts of generative models when the study treats them as such. The first public version had to be dated between

15 September (the release of JEV) and 1 October 2026 inclusive. Any language and any study design were eligible; non-empirical items (essays, position papers, methods notes) were included but are not counted in the evidence tallies. Papers that mention typed models without measuring or building one were excluded and, where useful, cited as context.

Grey literature. Blogs, technical reports and replication repositories were included when they report original measurements of a typed model, and are marked [†]. They were not searched systematically: they are the items surfaced by the two community trackers and by the web searches below. Vendor documentation is cited as context and is not counted as evidence.

3.3 Information sources, search and selection

We searched (i) arXiv, through 18 queries to its API (Supplement S5.3), on 1 October and again on 4 October 2026, the second run to catch late announcements, although the update search below found papers dated within the window that neither run had retrieved; (ii) Zenodo, through 11 queries to its REST API on 1 October 2026; (iii) two community trackers, “All about JEV” (2,073 records, 52 flagged as papers) [118] and *Awesome System One Models* (213 registry-verified papers) [119]; and (iv) Hugging Face Papers, OpenReview, SSRN and general web search. The arXiv search of 4 October returned 955 records (606 unique), 89 of them dated on or after the release, and the Zenodo queries of 1 October about 330 records, of which 62 met the keyword and date criteria. After screening by title, abstract and full text (flow diagram and counts in Supplement S5.4), the corpus contains 79 arXiv papers, 27 Zenodo records, one paper available only on alphaXiv and 7 grey-literature sources, that is, 107 studies and 7 grey sources. Of these, 85 report results that bear on at least one requirement and are coded below (Figure 1); seven more are non-empirical (essays, notes and one conceptual paper) and are recorded and appraised but not tallied, and the remaining 22 are background-tier engineering papers that are not coded. Excluded records with reasons are listed in `data/excluded_records.csv`; for the Zenodo screening of 1 October only counts were recorded.

Update search. To check whether evidence posted after the window changes any answer, we re-ran the searches on 8 October 2026 under a written protocol (`paper/review/UPDATE_SEARCH_2026-10-08.md`). The arXiv API rate-limited the requests, so the same 18 queries were run through arXiv’s advanced-search web interface (`scripts/rerun_arxiv_search_web.py`); each query read one results page of up to 200 records, and the largest returned 91 (per-query counts in `data/search_rerun_2026-10-08_query_hits.csv`) so no query was truncated, although the web interface and the API are different search backends and need not return the same records. The 11 Zenodo queries were re-run through the Zenodo API (`scripts/rerun_zenodo_search.py`), and the two trackers had added no qualifying item. We read 62 sources in full; 5 of them (1 arXiv, 4 Zenodo) measured no typed model and were excluded, and the remaining 57 (31 arXiv, 26 Zenodo; no new grey item) were appraised: 48 are coded (27 arXiv, 21 Zenodo), six are context, two are non-empirical, and one was excluded during coding because it evaluates no typed model. Of the coded sources, 30 were first posted 2–8 October; 5 are arXiv papers dated within the window that the 1 and 4 October searches had not retrieved (we call them late-indexed, but cannot tell indexing delay apart from the change of search backend); and 13 are Zenodo records dated within the window, a third set added after coding had begun (the protocol named only the first two). Because the 1 October Zenodo screening recorded exclusions only as counts, we cannot tell whether these 13 were excluded then or missed; they were re-screened under the same criteria, and the gap is a limitation of that screening. Together, 18 of the 103 coded studies dated within the window (17%) were found only by the update search, so the main search’s recall was

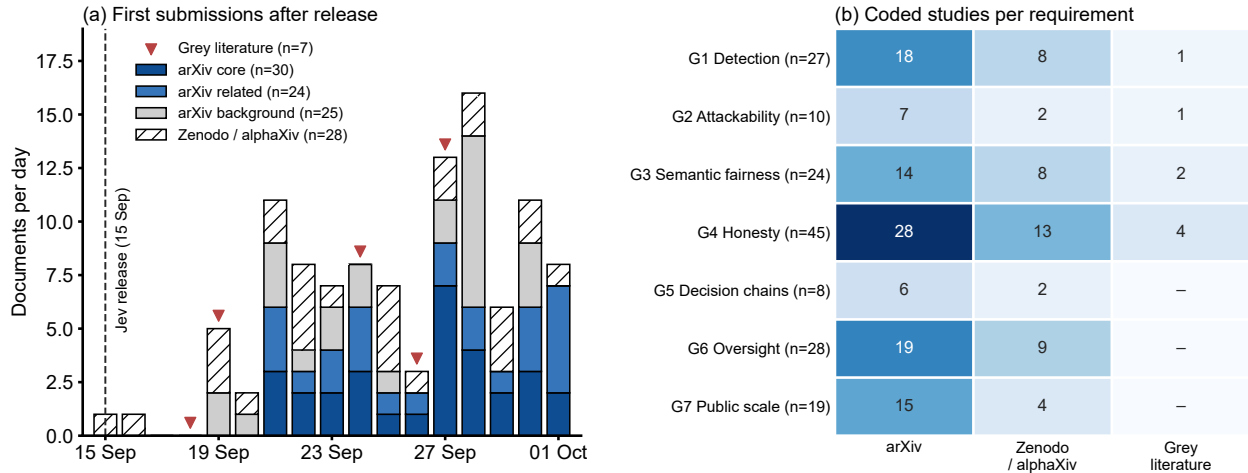


Figure 1: The evidence base. (a) First submissions per day after JEV’s release: arXiv papers by tier, Zenodo and alphaXiv records hatched; triangles mark grey-literature sources. (b) Coded study–requirement pairs that enter the tallies, by requirement and source; the middle column has 44 Zenodo and 2 alphaXiv pairs, and a study can inform several requirements.

incomplete; as the protocol required, they are not added to the main tallies, and Section 4.9 checks whether adding them changes the direction of any requirement. The 118 study–requirement pairs were coded with the same codebook and design rating and are reported separately (Section 4.9); the main-window counts, figures and tallies are unchanged. The protocol also froze the main-window certainty ratings, to change only if the update evidence changed a finding’s direction or rating; that freeze was not kept, because the certainty rule was revised during review, after the update studies had been coded, and every rating was re-derived (Section 3.5), a deviation from the update protocol. Unlike the main sample, every update pair was coded blind by a second AI coder; agreement was 97 of 118 ($\kappa = 0.55$, bootstrap 95% CI 0.36–0.70). This is a census of all update pairs, and its interval overlaps that of the main sample’s κ (0.42–0.94, from 30 sampled pairs), so the two values do not establish a difference; the census is the better estimate of how reproducible the codebook is. The 21 disagreements were resolved by a third AI pass that had read both rationales, which sided with the first coder in 11 and the second in 10, so the first coder’s code was overturned on 10 of the 118 pairs (8%; `data/coder_agreement_update_2026-10-08.csv`); main-window codes outside the 30-pair sample were coded once, and a similar share of them may be wrong.

The released codebook file (`paper/review/CODEBOOK_v3.md`) lacked, until 8 October, two rules that this paper states: rule (iv) of Supplement S5, that conditions constituting a requirement’s test are not added conditions (in Supplement S5 since the second review round), and the independence condition of the stronger-design rating. The main primary coders and the main blind second coder ($\kappa = 0.72$) worked from that file without the text of rule (iv); the update primary coders were told that the file contained it, although it did not; and the update second coders and the adjudicator applied it as stated in this section and in the corrected file. An audit of all main-window G2, G3 and G6 codes against rule (iv) in the twelfth review round recoded two G3 codes from Q to N [23, 120] (G3 from S 2, Q 12, N 10 to S 2, Q 10, N 12), which leaves the G3 answer, “not by default”, unchanged. Design ratings were always computed by script with the independence condition, so the second gap did not affect them.

3.4 Data extraction and coding

For every study we extracted the system actually measured (hosted JEV, an open model, or both), the model version, the task, the data and its language, the sample size, the comparators, the main results with a locator (section, table or figure) and the authors’ own caveats. Each arXiv or alphaXiv study was assigned a tier (*core*: bears directly on a requirement; *related*: supplementary evidence; *background*: engineering or other domains, coded only where it reports a measurement that bears on a requirement). Each pair of a study and a requirement it informs was then coded with a symmetric codebook (Supplement S5), always for *detector* use as defined in Section 1: *supports* (S) if the primary result shows the typed model performing the function at least as well as its comparators, or meeting the study’s own criterion, without an extra condition the study had to add; *negative* (N) if it shows the model failing the function or doing clearly worse than its comparators; *qualifies* (Q) if the result is mixed or depends on an added condition such as a locally fitted threshold, recalibration, a restriction of language or domain, or human review. Conditions that constitute a requirement’s test, such as adversarial content for G2 or name and order perturbations for G3, are not added conditions. For each coded pair we also recorded how the typed model compared with any generative comparator on that requirement (better, similar, worse, mixed or none). Because studies differ in tasks, metrics and designs, we do not pool effects; the synthesis is structured and narrative, with vote counts of the direction of each study’s primary result, as the SWiM guideline describes [121]. The codes, rationales with locators and comparator fields are released in `data/evidence_coding.csv` and listed in Supplement S2; codes count studies and are not effect sizes.

Agreement. A second coder coded a random sample of 30 study–requirement pairs blind to the first coding, from the full texts and the codebook alone. Agreement on the S/Q/N code was 26 of 30 (Cohen’s $\kappa = 0.72$, bootstrap 95% CI 0.42–0.94); the first coder’s codes in the sample were S 3, Q 23, N 4, so the sample says more about the boundary between Q and the other codes than between S and N. Both coders were AI agents of the same model family, which may inflate agreement. The four disagreements were resolved by re-reading the source, and one further code was revised after review; all resolutions are recorded in `data/coder_agreement.csv` and `data/evidence_coding.csv`.

3.5 Quality appraisal and certainty

Each coded study was appraised on three items: independence (independent, authors evaluating a system they built, or vendor-affiliated); design (preregistered; held-out evaluation, cross-validation or repeated runs; single run; case study; non-empirical); and whether sample sizes and uncertainty were reported. A study is rated *stronger design* if it was preregistered or held-out, reported sample sizes with intervals or tests, *and* evaluated a system its authors did not build; *weaker design* if it was a single run, a case study or non-empirical, or reported neither sample sizes nor uncertainty; and *moderate* otherwise, for example a held-out study that reported sample sizes without intervals. Of the 92 appraised sources, 37 are stronger, 17 moderate and 38 weaker (`data/quality_appraisal.csv`). None is peer reviewed, which applies to all equally and is therefore not a rating input.

For each finding carried into the abstract or the conclusion we rated the certainty of the evidence in the spirit of GRADE [122]. GRADE starts non-randomised evidence at *low*; benchmark evaluations are non-randomised, and we keep that starting point. The downgrading and upgrading rules below are our adaptation for single-model benchmark studies, not part of GRADE itself. A finding is *very low* if it rests on a single study, on studies none of which is of stronger design (two moderate studies, for example), or, when the claim concerns hosted JEV or typed models generally, on open-model evidence only; it is *low* if at least two studies agree in direction and at least one is of

stronger design; and it rises to *moderate* if at least three stronger-design studies, independent of one another, agree in direction and at least one of them is preregistered or independently replicated. A finding with several clauses takes the lowest rating of its clauses. A study counts as an *independent replication* only if its authors differ from the original’s, it uses its own data or a reimplementation rather than re-running the original code on the original data, and its design is at least moderate (a re-run of the original code on the original data is a reproduction); studies sharing an author count as one study in these counts. Each finding has a scope, shown in Table 3: hosted JEV, open models, or typed models generally (both). Evidence is counted per system arm: a study that tested hosted JEV and open models has a hosted arm and an open arm, and whether it supports or opposes a finding is judged on the result of the arm that is counted. For a finding about hosted JEV or about typed models generally, the rating is computed from hosted-model evidence alone, that is, from studies of hosted JEV and the hosted arms of mixed studies; open-model evidence (open-model studies and the open arms of mixed studies) can corroborate such a finding but cannot raise, lower or contest its rating. For a finding about open models, the rating is computed from open-model evidence. The rules also work downwards. A study *opposes* a finding when it reports, for the system and setting the finding names, a result on the outcome the finding states that runs the other way; a study with results in both directions on that outcome counts on the side of its primary result, and a result outside the claim’s scope, such as another system than the claim names, another outcome or a setting the claim excludes, neither supports nor opposes. Studies are ranked by design rating (stronger above moderate above weaker) and, within a rating, preregistered or replicated studies above the others. Any opposing study marks the finding *contested*; if the strongest opposing study ranks at least as high as the strongest supporting one, the rating also falls one level. For findings about escalation, a second stage counts as a *stronger reasoner* when, in the same study, it was more accurate than the first stage on the task on which escalation was tested; a difference within sampling error, or accuracy measures that point in opposite directions, makes it a peer. We applied these rules to every finding with all coded studies that bear on it, not only those cited for it. Every study coded on a finding’s requirements is listed, per system arm, as supporting, opposing, corroborating or out of scope, with a reason, in `data/certainty_evidence.csv`; the ratings are computed from that file and the design ratings by `scripts/rate_certainty.py`, which also checks that the ratings printed in Tables 3 and S8 match its output. The last rated row of Table 3, a tally of comparator codes rather than a set of studies, is descriptive: the rule does not apply to it, and we rate it low by judgement because the comparator field was coded once and its agreement was not measured.

These rules are not the ones first applied. On 4 October the ratings counted only the cited studies, and the rule could lower no rating. The rule was revised during review rounds 12–15, after the update studies had been coded, in response to reviewer findings that it worked only upwards, did not define an opposing study, and counted only the cited studies, and every finding was then re-rated with the revised rule on all coded main-window and update studies. Because the update protocol had frozen the main-window ratings (Section 3.3), this is a deviation from it, and Table 3 shows both the 4 October and the current rating. The 4 October column shows the ratings as given that day. Recomputed with the current rule on the studies cited that day, one of them would have been very low (note *a*): that thresholds set in advance miss error targets on new data was given as moderate because two of the four studies cited for it, including the only preregistered one, tested open models, and a third met its target on one dataset and missed it on another (Supplement S4.1). Against the ratings as given, one finding rises from very low to low, one falls from moderate to low, and four are newly contested; no finding reaches moderate certainty, and the other ratings are unchanged (Table S8).

3.6 Reviewers and tools

This analysis was carried out by one author with extensive use of AI agents (Claude Opus 5.5, Anthropic, for all agent roles). AI agents ran the searches, screened titles and abstracts, extracted data from full texts with locators, proposed and applied the codes, appraised the studies, served as the blind second coder, and checked the numbers in the text against their sources in separate review passes by fresh agents of the same model; the instructions given to the agents, their findings and the resolutions are released with the data. The author defined the research question and the requirements, set the codebook and the decision rules, and is responsible for every judgement in the paper; the author did not re-code or re-check individual codes by hand before this release. No human screened, extracted or coded in duplicate, so the Cochrane recommendation of dual human screening of at least a fifth of records [117] was not met.

3.7 Changes from the earlier release

No protocol was written or registered in advance. An earlier, abstract-based dataset release [13] used no codebook; this analysis replaced it with full-text reading, the symmetric codebook, a blind second coder, appraisal and certainty ratings, and a re-run of the arXiv search, and reclassified one paper that measures no typed model as context.

4 Evidence by requirement

Each subsection opens with a short answer and the tally of codes for that requirement, describes the studies that drive it, and relates the finding to earlier work; recommendations are collected in Section 6. “JEV” means the hosted model; its version is `jev-1.13` where a study reports one, and we flag studies that report none. Open models are named. Sources marked [‡] are Zenodo records and sources marked [†] are grey literature; all were read in full. Figure 2 summarises the codes, Table 3 the findings and their certainty, and Supplement S2 lists every coded study.

4.1 G1 — Can a typed safeguard detect violations?

Answer in brief

Yes as a ranking, not yet as a decision. On English benchmarks typed models rank harmful or misaligned content well at a small fraction of the cost of generative evaluators, better than low-cost evaluators but often below the best ones; thresholds fitted in one setting fail in another, and errors concentrate in sub-types that aggregate scores hide. Codes: S 2, Q 20, N 5; against generative comparators the typed models were better or similar in 6 of 21 pairs, mixed in 10 and worse in 5.

Zero-shot ranking. *Just Ask Jev* [16] benchmarked JEV (`jev-1.13.0`) on ten kinds of alignment failure across 44 benchmarks and 7,193 instances. A single generic yes/no question reached a median AUROC (area under the receiver-operating-characteristic curve) of 0.886 (95% confidence interval, CI, 0.821–0.952) over the 31 benchmarks with both classes present, without training, and beat an in-domain TF-IDF classifier and a response-length baseline on 25 of them; on the 19 benchmarks scored by an API LLM evaluator it cost \$0.30 against \$18.96 per pass. Tailoring the question to each benchmark changed the median AUROC by only +0.006 (CI −0.004 to +0.015); what the model is *shown* matters more, because many failures are defined against a reference such as the user’s stated belief or an injected instruction. In a matched comparison of hosted JEV, seven open

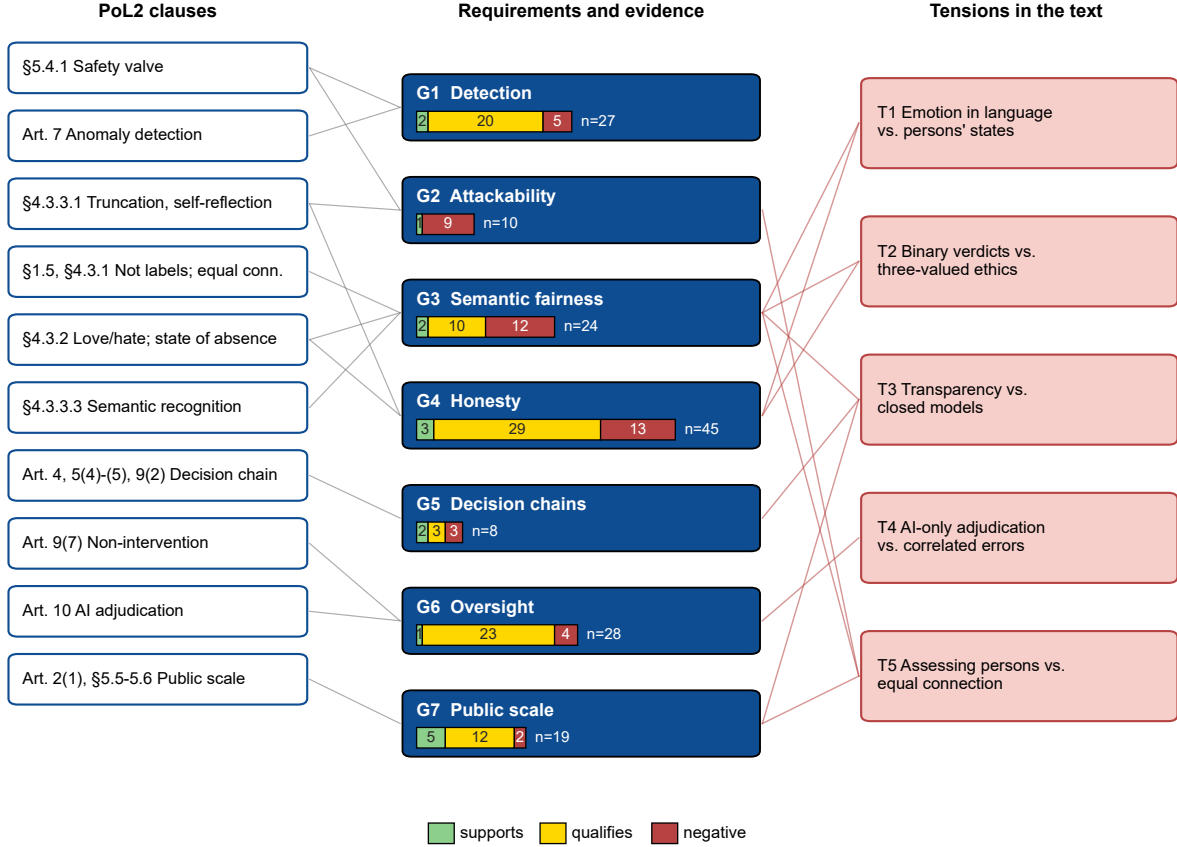


Figure 2: From clauses to tensions. Left: PoL2 clauses that require or constrain machine judgement. Middle: the seven requirements and the codes of the studies informing each, for detector use (supports, qualifies, negative; Supplement S5); bars count studies and do not measure effect size or certainty, and non-empirical items are excluded. Right: the five tensions of Section 5, linked to the requirements whose evidence they cite.

typed models, trained classifiers and LLMs, JEV was the most accurate typed model and separated out-of-scope requests with AUROC 0.921, but small classifiers fine-tuned on the labels beat it on intent detection (0.940–0.948 against 0.913) [31]. On shared samples of seven public moderation, hallucination, inference and injection tasks, JEV had the highest AUROC of the low-cost generative evaluators on all six binary tasks, although a frontier model costing 42–97 times as much beat it on two of the seven [123][‡]. Against the strongest generative evaluators it was often less accurate: it trailed the best LLM on 14 of 15 preregistered social-science annotation tasks by a median of 11.6 macro-F1 points [19], and on contract inference its accuracy (77.38%) was below the point estimates of all seven hosted LLMs, though not significantly below every one, at the lowest cost [124]. On agent trajectories it reached a positive-class F1 (harmonic mean of precision and recall) of 77.8 against 74.1 for the strongest generative evaluator, with lower precision and wins on two of four benchmarks (version not reported) [125].

Thresholds do not transfer. In *Just Ask Jev*, median F1 rose from 0.706 at the default threshold of 0.5 to 0.822 at a cross-validated one; the gain came mostly from benchmarks with rule- or evaluator-scored labels, and on human-validated labels the default did as well (0.721 against 0.694) [16]. A reproduction by another team, re-running the authors’ code and data on the `jev-latest` alias, covered five of its 44 benchmarks (median absolute difference 0.0036 from the paper) and found

the extreme case: on TensorTrust hijacking [61], AUROC was 0.95 but F1 was 0.158 at 0.5 and 0.947 at a threshold of 0.35 chosen by two-fold cross-validation; none of its five fitted thresholds was 0.5 [126][†] (Figure 3a). In a semantic-database workload, JEV at a fixed 0.5 reached a mean F1 of 63–66%, ranging from 24% to 89% by predicate [127]. Fitting to one domain can cost another: an open Laya guardrail fine-tuned on Chinese compliance queries cut its benign false-positive rate from 35.71% to 0.42% on its own test split but lost accuracy on a general Chinese safety benchmark (60.33% to 56.81%) unless earlier data were replayed during fine-tuning [128].

Aggregates hide concentrated failures. In agent-security screening, JEV was the strongest typed model on two of three tasks (interaction-risk AUROC 0.961 against 0.533 for Laya), yet it missed *all* 130 prompt injections that carried no explicit instruction, giving each an unsafe probability below 0.1; 130 of its 167 misses on that benchmark were made at confidence ≥ 0.95 [129]. An open Laya evaluator was near-perfect when an answer was stated explicitly but fell to 0.31 when it had to notice that something was *absent*, while a generative evaluator scored 1.00 on the same items [130][‡]. In police crash narratives, JEV found documented pregnancies almost perfectly within a regex-flagged stratum, but in the low-prevalence stratum only 14 of 42 adjudicated positives were confirmed, so the error rate measured in one stratum did not carry to another [131].

Where the evidence is not text. On numeric network-flow features, zero-shot JEV scored below always predicting the majority class (9.8% against 22.8%) and, even with 150 labelled examples in its context, stayed far below small tree models trained on the labels [132]. On a dated intrusion-detection benchmark with one labelled example per class, it reached F1 0.856, slightly below a generative comparator (0.880), with about 43 false alarms per run against about 764 for a random forest given the same single example per class (with eight examples the forest’s F1 was slightly higher) [133]. On a water-network screen it matched an expert rule tree (macro-F1 0.63 against 0.59) and beat a supervised classifier on event subtypes held out from its training, but stayed below fully supervised models and two frontier LLMs [134].

Relation to prior work. These results reproduce for typed models what was known for probability-reading safety classifiers [39, 40]: good ranking, poorly transferable thresholds and failures concentrated in sub-types [64]. Static scores also age: in English, replacing one word with a recently coined term of the same meaning flipped the hate-speech label of more than 10% of sentence pairs for 6 of 20 zero-shot models, and most models detected hateful posts less well in later years [135]. What is new is the price: ranking every message is now cheap enough to be a default.

4.2 G2 — Can the content under review steer the safeguard?

Answer in brief

Yes. Every study that attacked a typed model with the content it was asked to judge found some success, and the attacks that work need not look like attacks: an inserted opinion, fluent background text or a lexical lure; the one study coded as supporting found that injection rarely selected the attacker’s target, a finding that other attacks on JEV contest (Table 3). Where other systems were attacked in the same studies they were also vulnerable. Codes: S 1, Q 0, N 9; this is the most consistently negative requirement.

Injection and inserted content. On 510 reconstructed InjecAgent cases, malicious content shifted JEV’s action probabilities but selected the attacker’s target in only 9 cases (1.8%); “ignore

previous instructions” markers *reduced* the effect, and an adaptive attacker using score feedback reached 3.5% on fresh calls (18 of 510; CI 0–10%) [136]. *JevOut* [137] used the model’s own probabilities to search, within 64 accepted queries, for short fluent context additions constrained, and checked by a separate model, to preserve the question and the correct answer; these redirected 312 of 508 initially correct JEV decisions (61.4%) to an attacker-chosen wrong option, while three other decision systems, including a typed interface built on Qwen3-1.7B, were redirected on 64.9–73.2% of their own correct decisions (Figure 3b). In *JevAdvBench* [138], one unverified observer opinion appended to the state flipped 12.1% of JEV’s decisions (CI 8.5–15.5), statistically tied with impersonating an authority, the strongest injected command, although the command had to be placed in the question while the opinion needed only the state; benign rewording of questions stayed within re-run noise. When the correct action requires an unstated inference, words in the state that echo an option act as lures: on misleading steps of a household-task benchmark JEV was right 36% of the time and chose the lure 56% of the time [139]. Irrelevant but related paragraphs lowered its question-answering accuracy from 0.84 to 0.75 while its confidence stayed at 0.83 [25].

Adaptive and disguised attacks. Three multi-turn adaptive attack agents run against one JEV-based due-diligence application (version not reported) succeeded, for the strongest, in 25 of 27 runs, on average at the fourth turn, and marking content as untrusted made no meaningful difference; the authors call this a directional study of one application [140][†]. In a preregistered water-network study, attacks designed to look like normal operation were the hardest class (recall 0.34), and on a look-alike set 21 of the 76 windows an agreement gate accepted as benign were attacks [134]. In Kev-0.8B, an open JEV-compatible model, one planted note in a multi-question call could change several governance answers at once [141][‡].

Relation to prior work. These are typed instances of prompt injection and adversarial optimisation [57, 58, 60]. The feature specific to typed models is that the probability vector which makes them useful also guides the attacker: probability feedback succeeded on 63.6% against 56.4% with labels only, though the attack also works without it [137].

4.3 G3 — Are semantic judgements stable and fair?

Answer in brief

Not by default. Decisions can follow the names of options rather than the definitions bound to them, depend on order and wording, and lean towards one culture’s values; neutral option identifiers and option rotation remove much of this. Robustness varies by model: hosted JEV was less affected than Laya but more than some larger open decoders. Codes: S 2, Q 10, N 12.

Option names can override definitions. *Type-Safe Is Not Error-Free* [15] kept question, state, option names and definition texts fixed and swapped only which definition is bound to which name. For Laya, swapping the definitions bound to **yes** and **no** changed 76.9% of answers, against 6.5% for 0 and 1, and its AUROC fell from 0.938 to 0.232, an inversion that a type check cannot detect; hosted JEV (version not reported) changed 32.5% of its answers, about 24 times its test–retest floor, and its AUROC fell from 0.815 to 0.581 (Figure 3c–d). Under a matched harness, the same swap against fixed rubrics flipped 50.5 of every 100 answers of JEV 1.13.0 (floor 1.3), while digit or random-string identifiers brought it near the floor; Laya flipped 91.3 and four larger open decoders 4.2–6.5 [31].

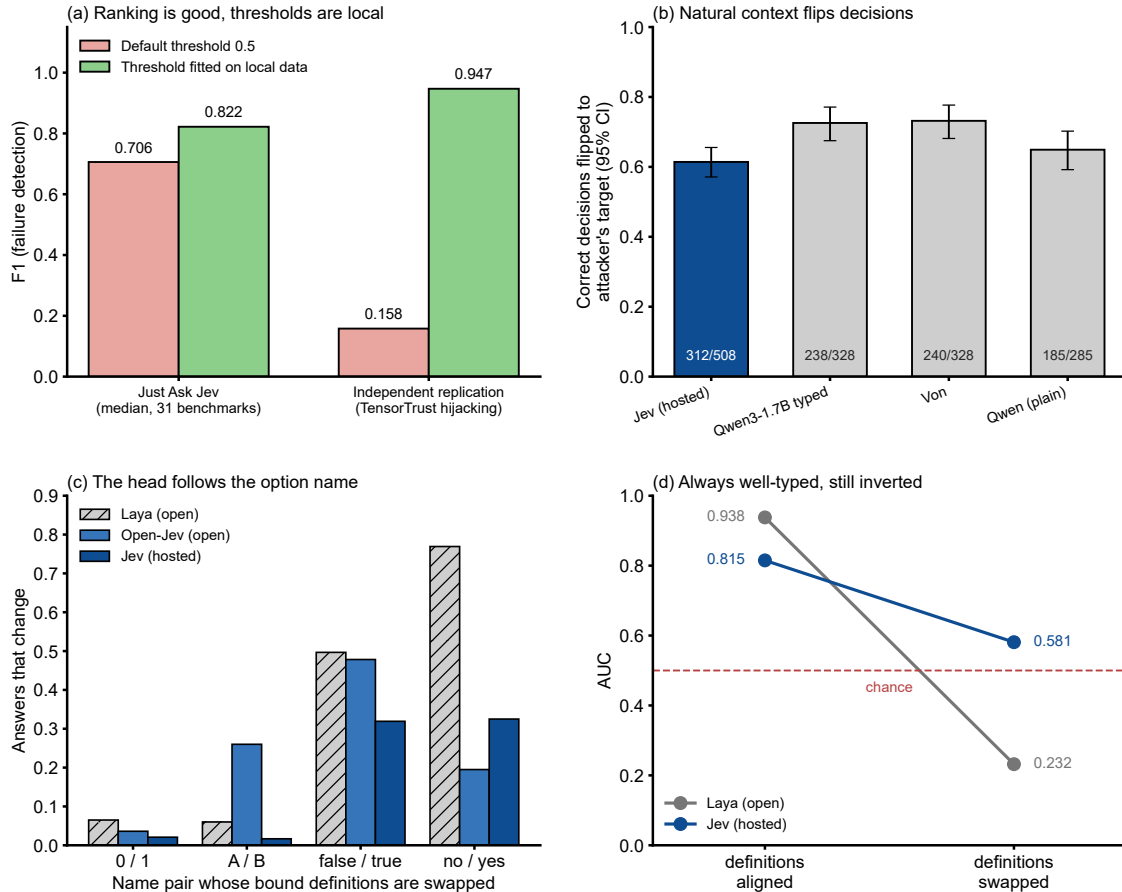


Figure 3: G1–G3: a detector, and a fragile one. (a) F1 at the default threshold versus a threshold fitted on local data [16, 126]. (b) Share of initially correct decisions redirected to an attacker-chosen wrong option by optimised fluent context additions, each system on its own correct items; “Von” is an open typed model evaluated in that study; error bars are Wilson 95% intervals from the reported counts [137]. (c) Share of answers that change when the definitions bound to a pair of option names are swapped, with question, state, names and definition texts unchanged; Open-Jev over 1,800 questions, Laya and JEV over 1,200 [15]. (d) AUROC with the *yes/no* definitions aligned with or swapped between the names; the type-error rate is 0% throughout [15]. Panels (a) and (b) use *jev-1.13.0* or the *jev-latest* alias; the study in (c)–(d) does not report the hosted version.

Order, wording and culture. On ANLI, JEV placed 38.8% of its predictions and 51.3% of its errors on the “neutral” option, although option positions were balanced, and on six datasets two random orderings of the same ordinal options agreed on 74.4% of items [142]. On 20-option tasks, reversing the option list changed about a third of the answers of typed readouts from open LLMs, and averaging over rotations reduced this to 14–18% [23]; for JEV, rotating the options left accuracy unchanged on the subjects one benchmark tested [143]. On moral dilemmas JEV was stable across runs but rewording moved its answers much more [144][†]. Posed as a forced choice between groups, “White” took 51% of JEV’s first picks across eight traits, although direct superiority claims were rejected with a typical “yes” probability of about 3%; this single informal test (version not reported) did not counterbalance option order [145][†]. Without a persona, JEV answered a cross-cultural values survey like its own American personas, and Saudi personas reproduced 87% of the human Saudi–US difference in English but 62% in Arabic [146]. Who made an argument mattered little to JEV in one study, but it was scored on a different rubric from the chat models, so we do not rank it against

them [147].

Language. TypeSafe states that languages other than English, including CJK scripts, are “handled but not equally well” [7]. A Chinese decision benchmark released as a repository found that hosted JEV needed no recalibration on Chinese voice routing and changed 1.9% of voice-routing decisions under option-order swaps (2.9% on customer service) and 1.7% between simplified and traditional script ($n = 323$), against 10.8% (voice routing) and 20.6% (customer service) option-order changes for the open Laya multilingual model [148][†]. On a second Chinese benchmark, built by the authors of Chinese models initialised from Laya, hosted JEV (version not reported) had a calibration error of 11.45% on general tasks against 3.78% for the authors’ general model, which slightly exceeded JEV in accuracy; their domain-fine-tuned specialists beat JEV in medicine but reached 92% of its average accuracy across specialised domains and were worse calibrated there (12.08% against 5.57%) [149]. In a preregistered test on Indian English dialect text, JEV’s calibration was the best of 22 models (ECE 0.063), on a single task [19]. All three models in a 37-dataset benchmark, typed and generative, degraded on low-resource languages [143].

Relation to prior work. Option-order and label sensitivity are long documented for LLMs and LLM evaluators [92, 94, 95, 99]; typed outputs do not remove it, and the name–definition inversion is a sharper form of it. The group and culture results echo the bias found in hate-speech classifiers [62, 63] and annotators [65].

4.4 G4 — Are probabilities honest, and is “insufficient” reported?

Answer in brief

Partly. Hosted JEV is often well calibrated on familiar closed-choice tasks, usually better than generative models’ stated confidence and mixed against frontier models; but calibration is local, related probabilities do not always cohere, an explicit “I don’t know” option is rarely chosen when it is correct, and “none” is chosen inconsistently. A separate yes/no question about whether the evidence suffices detects missing information better than confidence (AUROC 0.95 against 0.85 in one study), though it filters answers less well. Codes: S 3, Q 29, N 13; against generative comparators the typed models were better in 9 of 18 pairs and worse in 2.

Calibrated where the task is familiar. The broadest benchmark in our corpus, 346,009 requests over 37 datasets, found JEV more accurate than an open 27B model on 27 datasets (11 with non-overlapping intervals) and well calibrated for Choice questions when pooled (expected calibration error, ECE, 0.028; unweighted mean 0.061), with confidence that supports selective answering [143]. A blog test on 3,066 word-in-context pairs found an ECE of 0.027 after the prompt was tuned to the dataset [150][†]. In a controlled audit of 575,442 calls, JEV’s smooth calibration error was 0.015–0.033 on familiar closed-choice tasks [25]. Its annotation confidence was better calibrated than the stated confidence of 16 of 19 LLMs, though three frontier models had lower median error, with overlapping intervals (0.066 against 0.157) [19]. Other frontier comparisons are more favourable to JEV: on one of three medical benchmarks its probabilities were better calibrated than a frontier reasoning model’s (ECE 0.063 against 0.146), with similar error on the other two [151], and on six public binary tasks its raw Brier score was the lowest of all systems tested, though after recalibration a frontier model’s was slightly lower (0.071 against 0.076) [123][†]. An authors-built medical typed model also had lower calibration error than a matched generative baseline [152].

Calibration is local. On 2,416 blind human labels for crash narratives, raw JEV probabilities were too high overall, overstating prevalence by about 80%, and too close to the middle; out-of-fold recalibration reduced pooled ECE from 0.0231 to 0.0069 [153]. On radiology sentences, recalibration on a hold-out reduced ECE from 0.1235 to 0.0077, and an open baseline improved similarly [154] (Figure 4c). Used zero-shot to judge whether another model’s answer is correct, JEV was not usable off the shelf (ECE 0.133) [155], and on the ChaosNLI items where annotators disagreed most, the ECE of its Choice probabilities rose to 0.339, against 0.076 on the most agreed quarter, in a preregistered study that also found it far better calibrated than a local generative baseline and its confidence still ranking contested items (AUROC 0.744) [156][‡]; with a prompt tuned to the dataset, a blog test found ECE 0.025 on 612 two-way α NLI items from ChaosNLI, scored against the crowd’s share over all items [150][†]. This is the general pattern of calibration under shift [82]; ECE values are not comparable across these studies.

Probabilities that do not add up. On 480 pairs “the label is X” and “the label is not X”, JEV’s two probabilities deviated from summing to one by 0.064 on average, five times its own repeat noise (Figure 4a), with deviations concentrated where it was unsure [157]. Its yes/no and Choice interfaces led to different act-or-defer decisions on 32.8% of 1,000 synthetic cases (a post-trained open model: 34.2%) [158]. A reconstruction of the vendor’s confidence formula from 738,164 answers implies that padding the option set inflates reported confidence [159][†].

The missing “insufficient”. Offered “I don’t know or cannot answer” on a medical benchmark, JEV chose it for 17 of the 162 items for which it was the correct answer (10.5%), similar to a frontier generative model (8.6%), and “None of the above” on 53.9% of the items where that was correct, against 73.9% for the same model [151]. When every listed answer was wrong and recognising this required arithmetic, it chose an explicit “None” on only 7% of items while answering 99% of answer-present items correctly; relabelling the option, or moving it on inventory cases, recovered almost nothing, whereas per-candidate yes/no verification reached 99%, a reasoning generative model rejected correctly, and a threshold on the probability of “None” chosen on development data raised correct rejection to 79% [160]. With no relevant information JEV still put up to 0.80 on a salient option; a separate question asking whether the evidence was enough separated complete from incomplete evidence on multi-hop questions with AUROC 0.95 against 0.85 for confidence, but as a filter for accepting answers it was 2–6 points less accurate than confidence [25]. In a small case study, JEV’s yes/no probabilities for conflict and for ignorance fell in the same band, while a Choice question naming both states separated them in all four cases [161][‡]. Narrow fine-tuning of an open readout suppressed “none of the above” and biased it towards “yes” while validation loss kept improving [24].

Relation to prior work. Locality of calibration and its failure under shift are established [76, 82, 83], as is the gap between confidence and abstention [86, 87]. New is the evidence that, for typed models, *asking* about sufficiency detects missing information better than reading confidence.

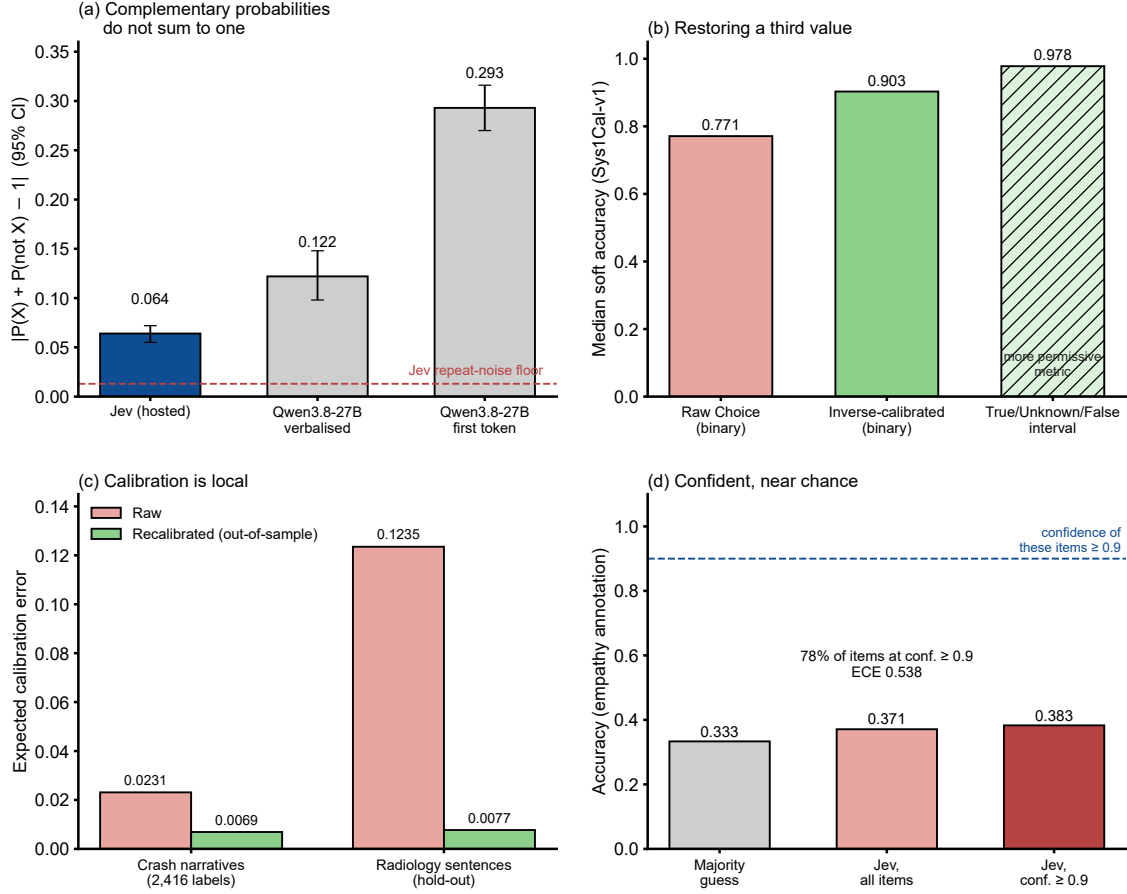


Figure 4: G4: honesty. (a) Mean absolute deviation of $P(X) + P(\neg X)$ from 1 over 480 negation pairs, against JEV’s repeat-noise floor [157]. (b) Median soft accuracy on Sys1Cal-v1 for raw Choice answers, after correcting for a suppressed “unknown” mass (a hypothesis of the study), and under a more permissive true/unknown/false interval metric (hatched) [162]. (c) ECE before and after recalibration on local labels, out of fold [153] or on a hold-out [154]. (d) Empathy annotation (three balanced classes): JEV’s accuracy on all items and on the items it answered with confidence ≥ 0.9 , against the majority base rate [19].

4.5 G5 — Can a decision that is only a number be audited and explained?

Answer in brief

It can be recorded and decomposed, not explained. A typed decision returns no reasons, and decisions assembled from several questions or interfaces need not cohere; writing intermediate facts into the state as separate steps makes chains inspectable and, when those facts are right, can make them more accurate, though in two later studies decomposition did not raise accuracy (Section 4.9). Codes: S 2, Q 3, N 3.

Records without reasons. A typed model returns a probability and nothing else, so nothing in its output shows which part of the input produced a verdict [163]. One Zenodo critique argues that removing reasoning traces makes responsibility attribution structurally harder [164][‡]; another argues that typed inference can relocate governance commitments into the schemas, option sets, rubrics and thresholds its developers write [165][‡]. When a solver was shown JEV’s probabilities while revising its logic-puzzle answers, it did no better than with a plain second look [166][‡].

Chains that do and do not compose. In ten scientific workflows, JEV made all intermediate semantic choices correctly, while seven wrong choices by comparator models, all on one question, changed downstream counts without changing the final labels: final-label accuracy alone could not reveal them [167]. Asked through two interfaces, the same decision led to different actions in 32.8% of cases [158], and category probabilities reconstructed from fine-grained answers differed from directly asked ones by a mean total variation of 0.22–0.35 [168]; per-step calibration audits can miss chain-level error when conditions shift [169][‡]. Decomposition can help: when JEV’s prediction of an opponent’s move was written into the state, its decisions in misleading games rose from 33% to 94% correct, but given a wrong written premise it followed the premise in 24 of 26 calls, which makes the written facts themselves the audit points [139].

Relation to prior work. Explanations generated after a decision may not reflect what produced it [112], and documentation standards [113, 114] describe models and data, not individual decisions; a typed safeguard needs a decision record and a separately sourced statement of grounds.

4.6 G6 — Does “act when confident, escalate when unsure” work?

Answer in brief

It depends on where escalation goes and how thresholds are set. Escalating uncertain cases to a stronger reasoning model, or accepting only verdicts an independently built rule confirms, saved a quarter to over half of the cost without loss of accuracy in several studies, a finding a later study contests (Section 4.9); escalating to peer models preserved confident errors, and thresholds fixed in advance missed their error targets on new data in most studies that set one, although in several, including a preregistered one, JEV’s threshold met its target (Table 3). Codes: S 1, Q 23, N 4.

Escalation that works. *JEV-as-a-Judge* [27] found JEV within three points of GPT-6 where a verdict can be read off the text, at 0.36% of its fee, and clearly behind where it must be derived; with a threshold frozen in advance a cascade beat GPT-6 by 0.93 points (CI 0.24–1.66) at 41.4% of its fee on 1,610 held-out pairs (83% from RewardBench; on JudgeBench it was 0.74 points below), and in a prospective live test on two new workloads, with thresholds set on 100 local labels each, it matched GPT-6 at 75.5% of the fee (Figure 5a). In a preregistered study with sealed test rounds on simulated network data, accepting a benign JEV verdict only when an independently built rule tree agreed let a frontier-LLM reviewer skip 34.5–38% of cases without loss of macro-F1, including on transfer to two other networks, although disguised attacks still passed (Section 4.2) [134]. A preregistered annotation cascade matched LLM-only quality at a quarter to half of the cost [19], and a semantic database with uncertainty-band escalation raised mean F1 from 63–66% to 96–98% [127].

Escalation among peers. In a rubric-grading study, JEV’s accuracy differed significantly from that of three low-cost generative evaluators in at most 8 of 27 paired comparisons, at 1/16 to 1/325 of their cost [170]. On JEV’s 12 most confident errors on each of seven ordinal grading panels (84 items), three low-cost generative evaluators gave JEV’s wrong answer in 242 of 252 verdicts (96.0%), against 50.3% expected if errors were independent; because these items were selected as JEV’s most confident errors, we read this as an upper bound on error correlation, though it is the set that matters for escalation [170] (Figure 5b). Replaying recorded verdicts, the best cascade’s average gain over the best single evaluator was at most 1.5 points, and on six of nine panels it was below it (Figure 5c). In agent-security screening, generative evaluators caught at most 5 of JEV’s 130

high-confidence misses [129]. In another study, escalating JEV’s uncertain band to a low-cost LLM never helped significantly; only a frontier model, at 42–97 times the cost, helped, on 2 of 30 cascades, so within the low-cost tier its author recommends routing that band to a person [123][‡].

Thresholds and gates. Under a 1% miss limit, symmetric allow/block/review policies automated nothing for JEV on prompt injection, and independent thresholds automated 47.03% of interaction-risk cases almost entirely through additional *blocks* [129]. In a preregistered annotation study, a confidence threshold of 0.9 fixed in advance was to give at least 0.85 accuracy; it reached a median of 0.815 and fell short on 8 of 14 tasks [19]. With thresholds chosen on held-out data, the one-sided upper bound on realised risk exceeded a 5% target for every model but one in a matched benchmark, while JEV’s realised risk met the target on one dataset (0.049) and missed it on the other (0.058), and a threshold set for in-scope requests let JEV accept 31.0% of out-of-scope ones (19.5% with an explicit out-of-scope option) [31]. Confidence-based deferral worked in-distribution for an authors-built medical typed model, whose area under the risk–coverage curve was lower than a matched generative baseline’s on three of four task families [152], and JEV’s own risk–coverage curves were monotone on six public binary tasks [123][‡]; but on medical benchmarks JEV’s selective prediction was worse than a frontier model’s on two of three [151]. A 0.8 confidence gate let 12.6% of attack-induced flips through, while an inserted opinion pushed 38.0% of confident answers below it [138]. A conformal gate built on Laya’s confidence returned risk 0.585 at a nominal 0.05 [171][‡], and a preregistered reproduction’s frozen Laya gate missed its error target (0.140 against 0.10) [172]. In network control, a fine-tuned open model’s gate set in advance did not carry over to changed questions: it acted wrongly on up to 0.801 of them and exceeded a 0.05 target on 6 of 7 changed templates, whereas JEV’s gate, which read the questions, stayed within the target on 6 of 7 and reached 0.143 on the seventh, within the study’s shift bound, and a conformal label floor implies that certifying action on a new question type needs at least 19 labels of it at a 5% error level [173]. Gates can also work: for Kev-0.8B, a threshold set by coverage rather than fixed at 0.8 sent every answer an attack flipped from right to wrong to review, although attacks reinforcing an existing mistake could pass [141, 174][‡].

Relation to prior work. Cascades and learned deferral are established [88, 89, 104], as is correlated error among models [101, 102]; risk-controlled thresholds [90, 91] assume exchangeability between calibration and deployment data, which the G2 and G4 evidence says will not hold under attack or drift. The new finding is that independence must be built: a differently constructed rule helped, a peer model did not.

4.7 G7 — Judgement and simulation at public scale

Answer in brief

Typed models make population-scale reading and simulation cheap, and some studies show how to do it honestly: audit samples, and carry the measured gap to real people into every conclusion. Simulated publics lean towards one population and understate disagreement, and deployment depends on one closed vendor. Codes: S 5, Q 12, N 2.

KITE [28] calls JEV once per unique state, simulates a million agents in under a second by table lookup, and propagates the measured human–model discrepancy as a shared error term into every conclusion; on 37 unseen studies this term gave retrospective coverage of 0.93 and 0.96 at nominal 80% and 90%, above nominal and possibly conservative, although its preregistered primary endpoint

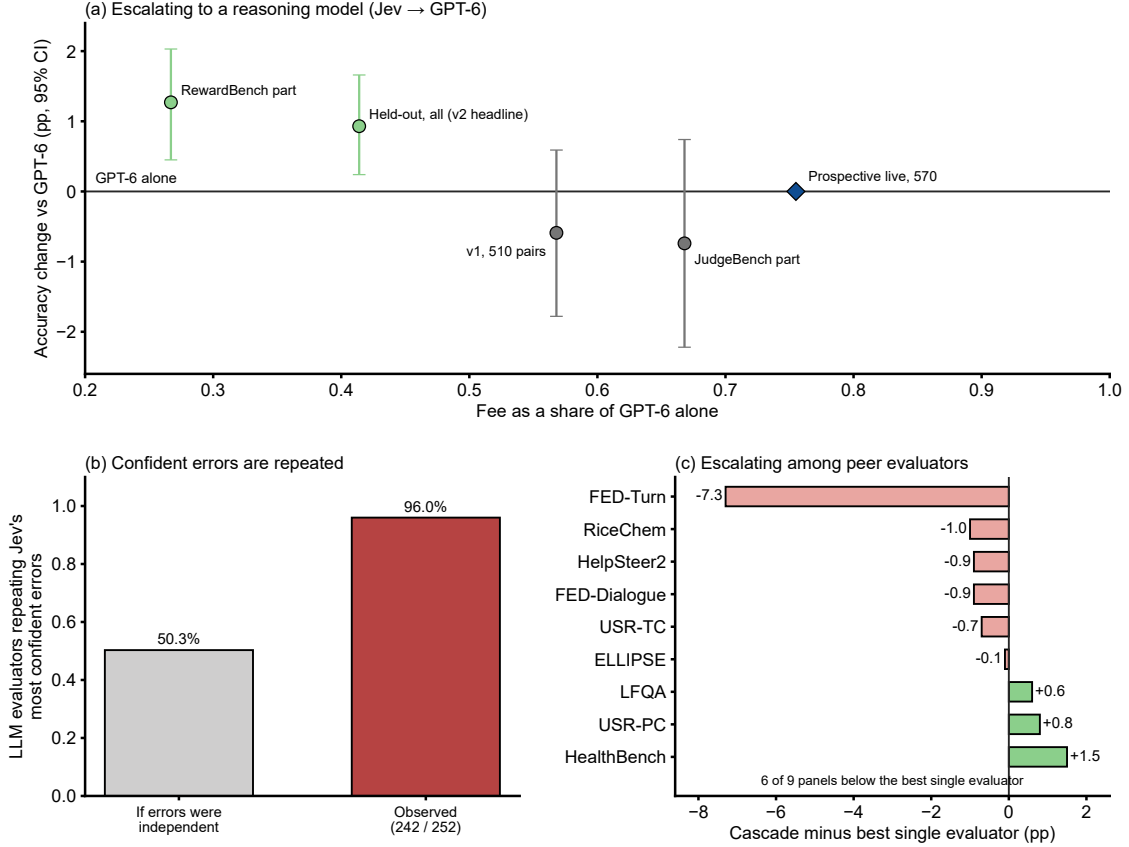


Figure 5: G6: escalation. (a) Accept-when-confident cascades from JEV to GPT-6: accuracy change against fee share [27]; grey points have intervals including zero, the diamond is the prospective live test. (b) Share of JEV’s most confident errors repeated by low-cost generative evaluators, against the share expected if errors were independent [170]. (c) Cross-fitted cascade minus best single evaluator on all nine panels [170].

showed no clear gain. A statewide study read about half a million police narratives with JEV, audited a probability-stratified sample by blind human adjudication, and reported misclassification-corrected counts with intervals, while finding residual personal identifiers after two redaction passes in 43% of the narratives documenting fetal harm and 37% of the adjudication sample [131]. Without a persona, JEV answered value questions like its American personas and stayed confident where annotators were split (Sections 4.3 and 4.4), so a simulated public leans towards one population and understates disagreement [146, 156][‡]. Matched benchmarks put hosted JEV at under a cent per thousand decisions with a median latency of 138 ms [31]; its weights are not published and it is not offered for self-hosting, which led one routing study to keep its interface classifier-agnostic with a self-hosted fallback [17]. A conceptual paper argues that cheaper evaluation may increase total evaluation and exception handling rather than reduce it [175]. The honesty conditions here are those of survey methodology and of disagreement-aware annotation [66, 67].

4.8 Across requirements

Of the 161 coded study–requirement pairs, 16 support detector use, 97 qualify it and 48 are negative (Figure 2). The shares change by about six points or less when only stronger-design studies are kept (7, 45 and 27 of 79) or when Zenodo records, which were negative more often than arXiv papers (19 of 44 pairs against 25 of 107), are dropped (13, 75 and 29 of 117). Seventy-six pairs compared

a typed model with a generative evaluator on the same requirement: the typed model did better in 23, similarly in 17, mixed in 28 and worse in 8. For failures specifically the picture is thinner and less favourable: of the 14 negative codes with a comparator, the typed model was worse in 6, similar in 4, mixed in 3 and better in 1, and 34 of the 48 negative codes had no comparator at all. The evidence therefore does not show that typed models are generally worse than generative evaluators, nor that their failures are shared with them; for most failures the comparison has not been made. Two failure modes follow from the typed interface itself: reported confidence can be inflated by padding the option set (G4), and a decision carries no reasons (G5); a third, that the probability vector guides attackers, is shown in one study (G2).

Read together, these codes support detector use only in the qualified sense used throughout: almost every success depended on an added condition, such as a locally fitted threshold, a second independent check or human review, and the failures in G2 and G4 are failures of the detector itself, because a steered or silent detector misses violations. The recommendation of Section 6 is therefore not that typed detectors are reliable, but that their failures are cheaper to contain in a detector role, where every consequential action passes another check, than in a judge role, where none does.

Table 3 lists the findings carried into Sections 5 and 6 and the abstract, with the certainty of each under the rules of Section 3.5, applied with every coded study of the main window and the update search that bears on a finding, and, for comparison, the rating as given on 4 October; no finding reaches moderate certainty, and four are contested.

Table 3: Key findings and the certainty of the evidence. *Scope:* hosted JEV, open models, or both (typed models generally); findings about hosted JEV or both are rated on hosted-model evidence alone, counted per system arm. *Codes:* the supporting studies’ codes. *Sources:* the principal supporting, then opposing studies (every study counted, with its direction, is listed in `data/certainty_evidence.csv`); “update:” update-search studies; “open:” open-model studies, which corroborate (or, “against”, point the other way) but do not enter the rating. Certainty as defined in Section 3.5; ^c contested within the scope; ^j descriptive tally rated by judgement. ^{4 Oct:} rating as given on 4 October (cited studies, earlier rule); ^a very low under the current rule (Supplement S4.1). *Now:* current rule, all coded main-window and update studies (main window alone: Table S8). Computed by `scripts/rate_certainty.py` from `data/certainty_evidence.csv`.

Finding	Scope	Codes	vs. generative	Sources	4 Oct	Now
Ranks harmful or misaligned content well zero-shot at a fraction of the cost; often less accurate than the best LLMs	hosted	S/Q/N	better (low-cost); worse (best)	[16, 19, 123, 124, 126, 129]	low	low
Fixed thresholds do not transfer; local fitting is needed	both	Q	–	[16, 19, 25, 27, 31, 126, 127]; opposing: [173]; update: [176–178]	low	low ^c
Errors concentrate in sub-types hidden by aggregates	hosted	Q	mixed	[127, 129, 179]; update: [180–183]	very low	low
Inserted opinions, optimised context and lexical lures steer decisions	both	N	similar	[25, 134, 137–139]; open: [141, 174]	low	low
Instruction injection rarely selects the attacker’s target (1.8% raw, 3.5% adaptive)	hosted	S	–	[136]; opposing: [138, 140]; update, opposing: [184]	very low	very low ^c
Option names can override definitions; neutral identifiers remove most of it	both	N	–	[15, 31]; update: [180]; open: [185]	low	low
Defaults to one culture’s values	hosted	N	–	[146]	very low	very low
Well calibrated on familiar closed-choice tasks, mixed against frontier models; calibration is local	hosted	S/Q	better or mixed	[19, 25, 123, 143, 151, 153, 154]	low	low
Explicit “I don’t know” rarely chosen when correct; “none” inconsistently	hosted	Q/N	similar or worse (frontier); worse (reasoning)	[25, 151, 160]; update: [176, 186]	low	low
A separate sufficiency question detects missing information, but filters answers less well than confidence	hosted	Q	–	[25]	very low	very low
Escalation to a stronger reasoner saves cost without loss of accuracy on readable judgements	both	Q	similar	[19, 27, 127]; open: [187]; update, opposing: [177]; open, against: [181, 188]	low	low ^c
Acceptance on agreement with an independently built rule lost no macro-F1, but fewer disguised attacks were flagged	both	Q/N	similar	[134]; update, open: [189]	very low	very low
Peer evaluators repeat JEV’s most confident errors (upper bound)	hosted	N	similar or mixed	[129, 170]	low	low
Thresholds set in advance miss error targets on new data	both	Q/N	–	[19, 129]; opposing: [173]; open: [171, 172, 188]; update: [176, 177]; update, opposing: [178, 190]	moderate ^a	low ^c
Typed models are not generally worse than generative evaluators; most failures lack a comparison	both	–	23 better, 8 worse of 76	Supplement S2	low	low ^j
No study measures the PoL2 construct (three-valued, Chinese)	–	–	–	–	–	–

4.9 Update search (2–8 October 2026)

The update search (Section 3.3) added 48 coded sources: 30 first posted 2–8 October, 5 late-indexed arXiv papers and 13 re-screened Zenodo records. Their 118 study–requirement pairs are coded S 12, Q 92 and N 14 (stronger-design studies only: 5, 45 and 9 of 59); they are not pooled into the counts above, and Table 4 sets them beside the main window. As a sensitivity check, we added the 18 records dated within the window (36 pairs: S 5, Q 28, N 3) to the main tallies; this changes no answer box, and the most frequent code changes only where Q and N were nearly tied in the main window (G3: N 12 against Q 10 becomes Q 14 against N 13; G5: a 3–3 tie becomes Q 4 against N 3). The check concerns the direction of the codes only; it does not recompute the comparator shares, and the certainty ratings of Table 3 are assessed with all update studies, including these 18, added. The update set is less often negative than the main window (12% of pairs against 30%): only one new study attacked a typed model (G2, the most negative requirement), and most new pairs are mixed. Of its 14 negative codes, 9 had a generative comparator and the typed model was worse in 6, four of them for open models only; across both sets, 39 of 62 negative codes still lack a comparator.

No answer box is reversed. The new evidence qualifies one statement, on decomposed chains in G5, and the box has been amended accordingly; the other boxes are unchanged, and the new evidence mostly confirms them. The certainty rule of Section 3.5 was revised during review, after these studies had been coded, and every finding was re-rated with it on all coded main-window and update studies, a deviation from the update protocol, which had frozen the main-window ratings. Against the main-window studies alone, the update studies change one rating: two stronger-design studies of hosted JEV raise the finding that thresholds set in advance miss error targets on new data from very low to low, still contested [176, 177], while two others whose thresholds met their targets add to the opposing evidence [178, 190][†]. They also make escalation to a stronger reasoner contested, because a JEV pre-screen cut a safety reviewer’s cost by only 4.3% [177], and add an opposing study to the injection finding, which the main window already contests. The derivation for each finding, and the changes against 4 October, are given in Supplement S4.1. Below we give, for each requirement, the studies that bear most directly on PoL2.

Table 4: Codes (S/Q/N) in the main window and in the update search, by requirement. *Update:* all update pairs; *in-window:* those from the 5 late-indexed arXiv papers and 13 re-screened Zenodo records, dated within the main window; *main + in-window* is the sensitivity check. Last column: update pairs with a generative (gen.) comparator on which the typed model did better, similarly, mixed or worse, and the number of pairs with no comparator. Data: `data/evidence_coding_update_2026-10-08.csv`.

Req.	Main window	Update	In-window	Main + in-window	Update vs. gen.; none
G1	2/20/5	2/17/3	1/9/0	3/29/5	1/4/7/3; 7
G2	1/0/9	0/0/1	0/0/0	1/0/9	–; 1
G3	2/10/12	5/15/3	1/4/1	3/14/13	3/2/3/0; 15
G4	3/29/13	3/26/3	2/6/1	5/35/14	6/0/7/3; 16
G5	2/3/3	0/6/0	0/1/0	2/4/3	1/1/2/0; 2
G6	1/23/4	1/19/3	1/7/1	2/30/5	3/3/2/0; 15
G7	5/12/2	1/9/1	0/1/0	5/13/2	1/0/6/1; 3
All	16/97/48	12/92/14	5/28/3	21/125/51	15/10/27/7; 59
Stronger design	7/45/27	5/45/9	–	–	–

G1: confirmed. Three new studies test hate speech and moderation, the core of the safety valve. Without task training, JEV’s binary hate-speech macro-F1 on four English datasets was 0.826, 0.691,

0.738 and 0.960, against 0.897, 0.682, 0.776 and 0.976 for GPT-6 Sol, and commercial LLMs were significantly ahead of every decision model on only one dataset; all models, typed and generative, struggled to separate hate from merely offensive language (three classes: JEV 0.532, Sol 0.604, a supervised TF-IDF classifier 0.639) [181]. This is closer to the best LLMs than “often below the best ones” suggests, but within the box’s direction. In online moderation, JEV’s F1 matched or exceeded Detoxify’s on two English datasets (though its accuracy on one was lower, 74.4% against 81.0%) and its AUROC on a Bluesky benchmark was 0.89–0.90, but its exact-match accuracy on Facebook hate-speech policy selection was 29.4% without retrieved precedents (51.6% with them), and on a community-rule test set it scored below always answering “no rule broken” (48.4% against 50.0%) [180]. On ToxicChat, a threshold chosen on 1,000 development labels gave 90.52% recall at a 3.24% false-positive rate on a test set of previously observed items with mixed labels, within the study’s 5% tolerance, against 35.6–58.9% recall for three fixed encoder conditions; in a post hoc breakdown, the false-positive rate on the human-annotated negatives alone was 5.39% [190][‡]. As a zero-shot detector of prompt injection, jailbreaks, harmful content and AI-generated text, hosted JEV reached AUROC 0.95–1.0 on eight benchmarks, but its accuracies (87–99% on six security benchmarks) were measured at thresholds fitted on held-out examples, and on AI-generated text accuracy rose from 78% at the default threshold to 89% only with such a threshold [184]; in an agent harness it screened injections well (AUROC 0.98) yet passed 25.5% of harmful requests and was unusable as a personal-data gate (false-positive rate 63–66%) [177]. On AgentHarm harmfulness judgements it was level with the best of nine generative configurations in a single run (84.54% against at most 85.57%) [191]. Aggregates again hid sub-type failures: human coders confirmed 96–100% of JEV’s crash-narrative flags where a coded field also recorded the factor, but 15% for unbelted occupants [183]; this study extends [153], by the same authors on reused data, and is not independent of it. Two open typed models scored below the generative comparators on the same benchmarks [188, 192], and in industrial fault detection hosted JEV flagged 57–94% of fault-free windows and, under nine shift tests, did no better than flagging every window [193].

Emotion and Chinese text (T1). Zero-shot on eight affective benchmarks, JEV averaged 62.93% macro-F1, above five of ten generative configurations and below the best (67.28%), at 96% lower cost than the best; on Chinese emotion classification (CMMA) every system was weak (JEV 21.06, generative models 19.82–23.31) [194]. This single-run study classifies the emotion expressed in texts, not persons’ states, so it bears on the side of the boundary that T1 would keep (Section 5): emotion scores for every message are now cheap, and on Chinese text they are not accurate. No new study manipulated language in Chinese; one engineering benchmark built from Chinese requests is discussed under G4 [195].

G2: confirmed. One new study attacked hosted JEV: a JEV-specific injection and a universal suffix reached 97–100% attack success on three tasks, and poisoning 5–10% of the training data implanted backdoors in an open model [184]. This attack succeeded far more often than the one in the main window’s supporting study (1.8% target selection) [136], which used a different attack and benchmark; that finding was already contested by other attacks on hosted JEV [138, 140] and stays at very low certainty.

G3: confirmed. Supplying each hate-speech dataset’s own definition changed 2.8–27.9% of decision-model predictions, with significant gains and losses equally frequent (8 each of 28 comparisons); JEV was the least sensitive, and a policy-trained generative model also lost agreement on three of four datasets [181]. Replacing opaque policy identifiers with readable hate-speech policy

names lowered JEV’s binary accuracy by 5.06 points (CI 3.91–6.21) [180], and in three open Laya checkpoints a label that contradicted its definition dropped accuracy to 0.14, below a no-information control [185]. Neutral identifiers are not free, however: in matched controls they cut an open generative readout’s detection of missing answers by 20–31 points [176]. Option order mattered much less for hosted JEV than for open typed models (1.8% of decisions changed against 30.3% for Laya [177]; order sensitivity 0.039 against 0.37–0.59 for Laya variants [196]), and JEV was more order-consistent than Qwen3-32B [197], consistent with “robustness varies by model”; but rewording definitions or rendering the same evidence differently still moved its answers [182, 193], and order flips concentrated where it was unsure [198]. An open model trained on Norwegian and Austrian register texts agreed with the reference on 94.6% of rule types in the Austrian test set but on 62.4% in an unseen Danish register [199][‡].

G4: confirmed. Comparisons with frontier models remain mixed. JEV’s ECE was lower than that of every generative configuration on a 10,938-question bilingual benchmark (0.053 against 0.078–0.253), yet it put 89% of the probability mass on the mode where the true share was 36% [191]; it was better calibrated than a frontier model’s token probabilities on eight human-labelled reading tasks, but a 27B open model was better calibrated on pairwise tasks [198]; on Chinese-language engineering questions its ECE was 0.027 against 0.031–0.061 for three flagship LLMs’ stated probabilities, although it failed 678 of 5,000 records [195]; and in game states it had generated itself it was worse than a constant predictor (Brier 0.283 against 0.250), where a small generative model read through option probabilities beat the constant [200][‡]. On offensive-language and hate-speech tasks its probabilities were poorly calibrated without retrieved precedents (ECE 14.9% on OLID, 33.2% on HateXplain; 9.7% and 20.2% with them) [180]. On abstention, with an explicit NONE option JEV rejected 63.7% of artificially omitted answers at 3.3% false rejection (an open generative readout: 45.0%) and 82.4% on a second task, but detection fell as candidates were added [176]; hosted JEV, not trained for the task, gave a hard answer to 62.8% of 382 unanswerable items, against 17.5% for the authors’ looped model [201]; and an explicit ABSTAIN option in an open JEV-style readout rejected 28 of 30 cases with no valid answer but chose correctly in only 17 of 30 valid ones [202]. Given a reject or escalate option, hosted JEV asked for new candidates in 45 of 48 contracts that had no valid one but escalated only 4 of 120 forwarding instances with no feasible path [186]. These fit the box: an explicit “insufficient” option helps only partly, and “none” is chosen inconsistently, for 3–94% of the items without an answer across studies of hosted JEV [160, 176, 186].

G5: one statement qualified. Two studies decomposed a judgement into criterion questions combined by a fixed rule: for hate speech this raised macro-F1 in 6 of 30 comparisons and lowered it in 11, and JEV’s one gain was matched by a tuned threshold [181]; on eight affective benchmarks it lowered JEV’s average from 62.93% to 57.31%, and appending the judgements to the state was not more accurate either (61.61%) [194]. In two others, exposing only the fields relevant to the current step, or moving checks into a separate gate, raised correct decisions (from 90 to 113 of 120; from 29 to 84 of 120) [186, 202]. Decomposition thus makes chains inspectable but more accurate only in some settings, and the G5 box now says so. A proof of concept recorded and reconstructed two live decisions, but only inside a deterministic shell its author built, and the output carried no reasons [203][‡], which confirms “recorded, not explained”.

G6: confirmed, less consistently. Thresholds fixed in advance often missed their targets on new data, though not always: rejection thresholds calibrated on one task exceeded a 5% false-rejection target in 6 of 12 transfers [176]; split conformal prediction reached its marginal target, although

JEV’s single-label answers under it were wrong 5.93% of the time at a nominal 5% [178][‡]; and abstention thresholds fitted on the training registers held a preregistered 2% target on the Austrian test set (1.3%) but gave 14.3% risk against a 5% target on an unseen register [199][‡]. Certification can also refuse: in the same study, a procedure whose bound is approximate rather than an exact finite-sample bound could certify no 5% automatic tier for JEV on AG News in any of 2,000 splits (191 of 2,000 on SST-2); an exploratory alternative threshold rule certified 1,776 of 2,000 SST-2 splits but only 4 on AG News [178][‡]. In moderation, accepting only JEV decisions with confidence ≥ 0.8 covered 71.9% of a Bluesky benchmark at 94.3% accuracy, but the accepted decisions were only 75.8% accurate on hate-speech policy selection and 67.6% on community rules [180]. Escalation to a stronger reasoner helped in some studies, matching a reasoning model at 30% of its calls [204] and, in an exploratory analysis, raising hate-speech macro-F1 when a frontier model (GPT-6 Sol) reviewed Laya’s least certain fifth, though not to the frontier model’s own level on two of three datasets, while referral to a cheaper model (GPT-5.6 Luna) lowered it on one [181], but not in others: a preregistered cascade from a near-chance open typed evaluator escalated almost every item [188], and a JEV pre-screen cut an LLM safety reviewer’s cost by only 4.3% once its own tokens were counted [177]. Escalating JEV’s least certain reranking labels to Qwen3-32B never helped [197]; on that task Qwen3-32B was about as accurate as JEV (agreement with human graders 87.2% against 86.2%, but a slightly lower reranker NDCG@10, 0.630 against 0.637), so it counts as a peer (Section 3.5), and the result is what the box expects for peers. Acting on a typed decision was safe where an independently built check held authority [202], and a gate fixed in advance with a rule-based fallback held on new data [205][‡]. The box stands, but the evidence that escalation saves cost without loss of accuracy is less consistent than in the main window: its certainty stays low, and the finding is now contested (Table 3).

G7: confirmed, with a new negative on privacy. On HateCheck, JEV cost \$0.033 per 1,000 texts at a median 336 ms, against \$1.177 and 1,426 ms for GPT-6 Sol, 1.6 macro-F1 points below it [181]; but on six of seven political-science tasks it was not cheaper than a frontier model at batch prices [198]. A two-tier probability sample of 229,510 crash narratives gave statewide counts that a blinded human tier confirmed, but only after per-factor human recalibration [183], the design the box describes. As a stand-in for crowd vote distributions, JEV’s Choice probabilities reported 0.88 on safety items on which people split 55 to 45 and did worse than a uniform guess [206][‡], confirming that it understates disagreement. New is evidence on privacy and dual use: conditional instructions in user input made hosted JEV’s labels alone reveal hidden gender, age-group, heart-disease and ethnicity attributes in every tested case, JEV served as an oracle that raised the judged harmfulness of harmful plans, and 200 of its labels trained a surrogate model from 49% to 92% agreement [184]. This extends the case for keeping probabilities from untrusted callers (Section 6) to the fields placed in the detector’s input and to the tasks untrusted callers may query.

5 Five tensions in the specification

Read against the evidence, the specification pulls in two directions at five points, and cheap typed judgement turns each from a philosophical question into an engineering default. For each tension we give the most charitable reading of the text, the evidence that bears on it, and options, at least one of which preserves the text. The choice among options belongs to the framework’s authors and to those who would implement it.

T1. Understanding emotion in language versus detecting persons’ emotional states.

§ 4.3 states that it is “entirely unnecessary” for PAI to detect a person’s emotional state, which is fleeting, personal, context-sensitive and touches on dignity; § 4.3.3.1 asks PAI to “understand human emotional states and processes”; and § 4.3.3.3 illustrates semantic recognition with two questions about a little dog whose difference lies in the intent of the utterance. Read together, the clauses admit a consistent interpretation: recognise the emotional meaning and intent of utterances, never infer a person’s inner state. The text does not state this boundary, however, and a typed model makes it almost free to attach an emotion score to every message, which is how the boundary would be crossed in practice. The evidence on emotion-related judgement is thin and mixed: in one annotation study JEV was confident on most empathy items yet on its most confident items barely more accurate than on all items (0.383 against 0.371, with three balanced classes), a failure the authors found in a low-cost generative model as well [19], while on multilingual valence–arousal regression JEV with a supervised layer fitted to the labels matched the best systems [207]. The wider literature is clearer: inferring emotion from observable signals is scientifically weak [72] and ethically fraught [73]. *Options:* state the boundary explicitly, so that a safeguard judges the semantics and intent of utterances but never assigns emotional states to persons; and reserve emotion-related judgement for PAI auditing its *own* outputs, for example for the fabricated intimacy that § 4.3.2 names as manipulation.

T2. Binary verdicts versus a three-valued ethics.

PoL2 is built on a love/hate polarity but adds the state of absence (§ 4.3.2) and warns that its concepts serve structural analysis, not moral labelling (§ 1.5). A yes/no question such as “is this hate language?” discards both qualifications. The evidence points the same way from several directions: option names can drive the verdict [15, 31]; complementary probabilities do not sum to one where the model is unsure [157]; an explicit “I don’t know” option is rarely chosen when correct, and “none” only inconsistently (54% of the time on a medical benchmark, 7% when recognising it needed arithmetic) [151, 160]; and a yes/no probability placed conflict and ignorance in the same band, whereas a Choice question naming both states separated them in four cases [161][‡]. It also shows what helps: a separate yes/no question asking whether the evidence suffices detected missing information better than confidence, although it was a weaker filter for accepting answers [25]. *Options:* make every love/hate question three-valued, with absence or insufficient information as an explicit option, and ask sufficiency as its own question whose use rate is monitored; keep “contested” (people disagree) as a separate flag routed to people, distinct from “absent”; and keep moral vocabulary out of option names by binding neutral identifiers to definitions and testing them by renaming. These options implement § 1.5 and § 4.3.2 rather than change them.

T3. Full transparency versus closed models.

Article 4 and Art. 5(4) require that operating logic and decision processes be “completely open and transparent” and that technology “should not be a black box”. TypeSafe has published neither the weights, the architecture, the reward nor the training data of hosted JEV, and does not offer it for self-hosting [17, 19, 31]. A charitable reading is that Art. 4 concerns the transparency of NaturalDAO’s processes and resource flows, which logging a closed component’s inputs, outputs, versions and calibration data partly satisfies; but such logs cannot show how the component reaches its outputs, and open weights alone would not provide the “reasoning steps” of Art. 5(5) either. Open models make the readout inspectable, but in several comparisons they were less accurate or less robust: Laya showed the most severe name inversion and was near chance on interaction-risk screening [15, 129]. The gap is not fixed: larger open decoders were more robust to option-name swaps than JEV [31], Chinese models initialised from

Laya slightly exceeded JEV on their authors’ general benchmark and, after domain fine-tuning, in medicine, though they reached 92% of JEV’s average accuracy across specialised domains [149], and an open RLCD recipe exists [208]. *Options:* (a) use closed models only as external baselines and run consequential judgements on self-hostable models whose weights, versions and calibration data are pinned in each decision record; or (b) allow closed models as one of several independent detectors, with logged inputs, outputs and versions. Option (a) trades measured accuracy and robustness for inspectability; that trade-off should be made explicitly and revisited as open models improve. Publishing decision records also conflicts with keeping probabilities from attackers (Section 4.2), which Section 6 resolves by delayed and coarsened publication.

T4. AI-only adjudication versus correlated errors. Article 2(1) makes NaturalDAO autonomous “on the premise of close collaboration with all humanity”, Art. 10 gives disputes to AI fact-finding and ethical assessment subject to human questioning, and Art. 9(7) states that human supervision “does not directly intervene” in decisions, with issues raised by questioning handled by NaturalDAO itself. The text does not specify how the premise of collaboration constrains individual decisions. The evidence bears on the non-intervention principle in two ways. Escalation can preserve errors: on rubric grading, low-cost generative evaluators repeated JEV’s most confident errors in 96% of cases (an upper bound, from items selected as confident errors), and evaluators caught at most 5 of 130 confident misses in agent-security screening [129, 170]. But escalation can also work, when the next stage is a stronger reasoner or an independently built rule [27, 134], although a later study contests the first (Section 4.9). Whether a human end-point would do better on governance items is untested: no study compares human reviewers with typed models on such items, and the oversight literature shows that people also over-rely on confident systems and disagree among themselves [106, 107, 109]. *Options:* (a) keep non-intervention, but make questioning effective by publishing a standing random audit of high-confidence decisions, with a duty for NaturalDAO to respond publicly to each discrepancy and to escalate to independently built checkers rather than peers; or (b) use the amendment procedure of Art. 11 to define a class of welfare decisions, such as withholding welfare (and, by the corresponding change to §5.4.1, blocking a person’s speech), whose automated execution a human reviewer can suspend pending review. Option (a) preserves the text but leaves consequential decisions with automated components, so it falls outside the detector role defined in Section 1; we list it because it preserves the text, not because the evidence supports it. Option (b) adds a check whose value is plausible from the due-process literature [3] and close to existing law (below), but untested on this task; the reference design of Section 6 implements it.

T5. Assessing each person versus equal connection. Among the things governed AI “can” do, §5.6 lists quantifying “each person’s Love Language and hate-language characteristics” and guiding behaviour with incentives. §4.3.1 holds that treating every person equally and treating every person’s behaviour equally “cannot be equated”, and the surrounding text refuses to label people by what they have done. A typed model can score every utterance of every person for a fraction of a cent, so a per-person aggregate is technically trivial. The same models answered value questions from a default cultural position [146], showed group bias in one informal forced-choice test [145][†], performed worse on low-resource languages (as typed and generative models alike did in one benchmark [143]), and could be steered by a third party’s opinion inserted into the content [138]. Aggregated over a person’s history, such errors could fall unevenly on groups and become a durable label that others can manipulate; no study yet measures differential error by speaker group on behavioural judgements. Experience with social scoring systems shows how such scores become infrastructure for control [55, 56]. *Options:* score events and behaviours, never persons, and keep

no persistent per-person aggregates; or, if § 5.6 is implemented, require consent, access to one’s own record, an appeal path, and published error rates by group and language for any per-person measure.

Regulatory parallels. None of these tensions is unique to PoL2, and law in the EU already takes positions on several, directly or by analogy. The EU AI Act [48] prohibits emotion recognition in workplaces and educational institutions except for medical or safety reasons (Art. 5(1)(f)); because the Act defines an emotion recognition system as one that infers emotions from biometric data (Art. 3(39); Recital 44), inference from text alone, which is what a typed model performs, falls outside the prohibition, and the provision bears on T1 by analogy rather than application; and social scoring, meaning the evaluation of persons over time on their social behaviour or personal characteristics where the score leads to detrimental treatment in unrelated contexts or to treatment that is unjustified or disproportionate (Art. 5(1)(c)), bearing on T5. For high-risk systems it requires effective human oversight (Art. 14), and it gives persons affected by decisions based on certain high-risk systems listed in its Annex III, with legal or similarly significant effects, a right to an explanation of the system’s role and of the main elements of the decision (Art. 86). These apply only to high-risk systems, and neither a speech filter nor a private organisation’s welfare decisions is listed in Annex III, whose point 5(a) covers decisions on public assistance taken by or for public authorities; point 8(a) does list AI used in alternative dispute resolution, which may reach the adjudication of disputes under Art. 10 where its outcomes produce legal effects for the parties (Recital 61). So for the safety valve and welfare decisions they bear on T4 and on the explanation duty of Art. 9(2) by analogy, whereas the GDPR applies wherever personal data are processed. The GDPR restricts decisions based solely on automated processing that produce legal or similarly significant effects and, where such decisions rest on a contract or explicit consent, requires at least the right to obtain human intervention, to express one’s view and to contest the decision (Art. 22(1), (3)) [51], which is close to option (b) of T4. For content moderation, the Digital Services Act requires hosting providers to give a statement of reasons for each restriction (Art. 17) and online platforms to run an internal complaint-handling system (Art. 20) [33], and the Santa Clara Principles ask for notice, reasons and appeal [34]; these are close analogues of a safety valve that blocks hate language. The options above, no emotional-state inference about persons, no persistent per-person scores, human-suspendable blocks with reasons and appeal, and published decision records, would bring an implementation closer to these norms.

Figure 2 shows which requirements each tension draws on.

6 A reference design for a clause-governed safeguard

The evidence suggests a plausible role for typed decision models in a safeguard such as the PoL2 safety valve: a fast, cheap, recorded *detector*, not a *judge*. Figure 6 sketches a design built on that principle for the valve of § 5.4.1; it applies to any governance layer that screens the inputs and outputs of AI systems. It is compatible with the pilot protocol of the NaturalDAO governance workstream, to which the author contributes and which already distinguishes **conforming**, **violating** and **insufficient** verdicts and the actions allow, repair, block, clarify and review [54]; the design adds a temporary *hold* and makes *block* final only after human confirmation. Where a recommendation coincides with a choice already made in that protocol we say so, so that readers can judge whether the evidence or the prior design drove it.

The design has seven elements.

1. **Give the detector the reference, not only the text.** Many failures are defined against

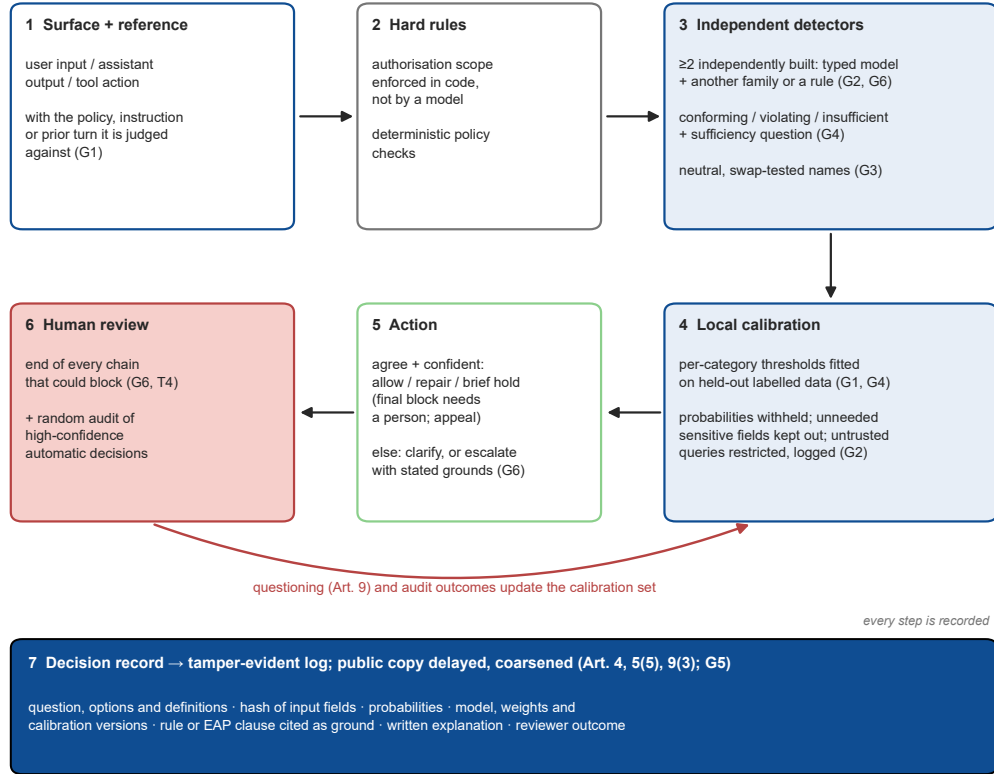


Figure 6: A reference design for a clause-governed safety valve. Typed models are detectors (3) whose output is calibrated locally (4) before any automatic action (5); disagreements and insufficient information are clarified or escalated, any final block is confirmed by a person, and a random sample of confident automatic decisions is audited by people (6); every step is written to a decision record (7). Labels in parentheses give the requirement or tension that motivates each element.

an instruction, a policy or a prior statement [16]; the content under review is always paired with that context. Facts that require computation or inference (sums, dates, consequences) are computed in code or in a separate logged step and written into the state, not left to the detector [139, 160].

2. **Enforce scope in code.** Whether an agent stays within its authorisation is a property of the surrounding system [209], enforced by capability checks before any model is consulted.
3. **Use at least two independently built detectors.** A blocking signal requires agreement between detectors that differ in construction, such as models of different families or a model and a rule, since peer models share confident errors [170], while agreement with a differently built rule let a reviewer skip about a third of cases without loss of macro-F1, although fewer disguised attacks were flagged as non-benign [134]. Every question is three-valued (conforming, violating, insufficient), sufficiency is also asked as its own yes/no question [25], and options carry neutral identifiers bound to definitions and pass a renaming test [15, 31]. Flags that must fail independently (allow/block, risk level, needs-human) are asked in separate calls, because in an open JEV-compatible model one planted note changed several answers in the same call [141][‡]. Self-hostable models are preferred where their accuracy suffices (T3). The three-valued verdict is already part of the workstream’s pilot protocol.
4. **Calibrate locally, keep probabilities from untrusted callers and restrict what**

they can query. Thresholds are fitted per category on held-out local data [16, 126, 153], with distribution-free risk control where its exchangeability assumption is plausible [90, 91], and re-certified for every new question type [173], on a schedule, and whenever a temporal re-validation fails (item 16 of Table 5). Raw probabilities are not returned to untrusted callers, because they help attackers optimise, although attacks also succeed with labels alone [136, 137]; fields a decision does not need, sensitive attributes above all, are kept out of the detector’s input, since a user authorised for the task, by planting conditional instructions in the input, recovered hidden attributes from labels alone with two to six queries per state; and queries from untrusted callers are restricted to authorised tasks, since labels refined harmful plans, and logged, since labels also trained a surrogate [184].

5. **Act automatically only on agreement, and restrict people only briefly and reversibly.** Agreement plus calibrated confidence allows automatic *allow*, automatic *repair* of the AI system’s own output (such as the truncation of § 4.3.3.1, never the editing of a person’s message), and a *hold* of a person’s content limited in time (for example, at most 24 hours) and fully reversible. A hold becomes a final block of a person’s speech, a benefit or an account only after confirmation by a person, and remains open to appeal. Disagreement or **insufficient** leads to clarification or to escalation to a stronger reasoning model that must state its grounds [27]. Because an automatic allow is consequential for anyone the allowed content harms, the miss rate is monitored and governed alongside the false-block rate. This element is what keeps the typed model a detector in the sense of Section 1.
6. **End every chain that could block with a person, and audit confident decisions.** Reviewers see every escalation that would end in a block, every unresolved escalation and a random sample of high-confidence automatic decisions, since the most damaging errors are confident and shared [129, 170]; their outcomes, and the results of public questioning under Art. 9, feed back into the calibration set.
7. **Record every decision.** Each record holds the question, option set and definitions, a hash of the input fields, the probabilities, the model, weight and calibration versions, the rule or clause cited as ground, any written explanation, and the reviewer’s outcome. Records are kept in a tamper-evident log open to supervisors (Art. 9(3)); the public version publishes probabilities coarsened and with a delay, so that it serves Art. 4 and Art. 5(5) without becoming an oracle for attackers.

Making human review workable. A person at the end of the chain is a safeguard only if the person can do the job. Reviewers need capacity: strict error limits shift volume into review and blocking [129], and inserted opinions push confident answers into review [138], so review load must be budgeted and reported alongside accuracy, and cheaper evaluation may increase rather than reduce the total [175]. Reviewers are prone to automation bias and complacency [106, 107] and can become the place where responsibility for system failures lands [110], so the model’s verdict and probability are hidden during the random audit, and agreement between reviewers, not only with the model, is measured. People also disagree on exactly the contested items where typed models are most over-confident [66, 68, 156][‡], so contested categories need more than one reviewer, and the person affected needs notice, reasons and a route of appeal [33, 34].

Trade-offs and known limits of this design. Requiring two detectors to agree before blocking lowers sensitivity: more violations are missed, which conflicts with strict miss limits that are otherwise met by blocking more [129]; the design accepts more holds and more review instead,

and its miss rate must be measured. Risk-control guarantees hold under exchangeability between calibration and deployment data; they bound error under benign drift, not under adaptive attack, which the G2 evidence shows is realistic. For hate speech the shift is not hypothetical: across 20 language models, rankings on four static English hate-speech benchmarks did not predict rankings on time-stamped or neologism-substituted data (Spearman’s ρ between -0.31 and 0.19 , all 90% intervals including zero), whereas the static benchmarks agreed with each other (mean $\rho = 0.36$) [135]. Separate calls for independent flags and a second detector multiply cost and latency, eroding the main advantage of typed models, although the measured costs leave a wide margin [31, 133]. Finally, most elements rest on evidence of low or very low certainty (Table 3); the design is a hypothesis to be tested on the PoL2 construct, not a validated system.

Table 5 turns the evidence into a checklist for any typed model proposed for such a safeguard. Items 1–4, 9, 14 and 15 draw on 9 of the 14 items of [12] (its B1–B2, C1, C3–C4, R1–R2 and T2–T3); the others are added for a governance safeguard, and that checklist’s B3, T1, T4, R3 and C2 are not adopted. Each item has a provisional pass criterion; the values are starting points to be set by those deploying the safeguard, not empirically validated thresholds.

Table 5: Evaluation checklist with provisional pass criteria for a typed decision model in a governance safeguard. “Basis” gives the evidence or clause; items marked ◦ are added beyond the checklist of [12].

#	Requirement	Provisional pass criterion	Basis
1	Compare with a label-probability readout of a generative model and a cheap generative cascade	Reported on the same items	[12]
2	Fit thresholds on held-out local data; report F1 at the fitted and the default threshold	Fitted threshold certified with a one-sided risk bound	[16, 31]
3	Report calibration per category, before and after local recalibration	Per-category ECE and slope reported with intervals	[153, 154]
4	Option-name swap test	Flip rate under a definition swap at most twice the repeat-noise floor	[15, 31]
5◦	Three options including <i>insufficient</i> , plus a separate sufficiency question	Recall of <i>insufficient</i> at least 0.5 on paired present/absent items	[25, 54, 160]
6◦	Coherence of complementary and exclusive questions	Deviation from summing to one within twice repeat noise	[157]
7◦	Red-team with adaptive, probability-guided additions, inserted opinions, lures and multi-turn attacks, and with label-only oracle attacks (attribute inference, surrogate extraction) under the deployed input and query restrictions	Attack success reported per attack type and below a target set in advance	[137–140, 184]
8◦	Errors disaggregated by sub-type, including confident errors	No sub-type with recall below a set floor	[129]
9	Decision-level stability over repeats, interfaces and decompositions	Disagreement across interfaces reported; one interface fixed per question	[124, 158, 168]
10◦	Error correlation with every escalation target	Repetition of confident errors below independence plus a set margin	[170]
11◦	Evaluation in Chinese and each language served, with script and order variants	Each language within a set margin of English	[7, 148]
12◦	No forced choices between groups of people; group-bias test on behavioural questions	Error rates by group within a set margin	[62, 145]
13◦	Value-laden questions: default position without persona, behaviour where people disagree	Reported; contested items routed to people	[146, 156]
14	Latency with and without caching; cost per correct decision	Reported, including review load	[12, 31]
15	Versions and inspectability	Model, weights and calibration versions pinned in each record; self-hostable or logged	Art. 4, Art. 5(4)–(5), § 5.5
16◦	Temporal re-validation: re-evaluate on a time-stamped hold-out of recent items and on paired items in which a target word is replaced by a newly emerged term of the same meaning	In each language served (item 11): per-period recall of hate language within a set margin of the certification period, and flip rate on neologism pairs within a set margin of the repeat-noise floor; when either is exceeded, thresholds re-certified by the procedure of item 2	[135] (English only; no typed model tested)

7 Discussion

7.1 Is hosted Jev a suitable governance tool for PoL2?

The preceding sections answer the research question in the terms of the detector–judge distinction. Here we state what that answer means for the specification itself, function by function, so that the verdict can be checked against the coded findings and their certainty (Table 3) rather than inferred. Table 6 takes each PoL2 function that a typed model could be asked to perform, names the requirements it draws on, gives the one or two findings that decide it, and assigns one of three verdicts: *not supported*, where the coded evidence shows the function failing or depends on a property the evidence contradicts; *supported only as a recorded detector with conditions*, where the function can be performed in the detector role of Section 1 inside the design of Section 6, and in no other way the evidence licenses; and *not tested on PoL2’s construct*, where no coded study measures what the clause asks for. Every certainty entry is the rating of the finding in Table 3; none is moderate, and a verdict that rests on a contested finding is marked as such. The verdicts concern hosted JEV at its versions of 15 September to 8 October 2026 and, where the finding’s scope is “both”, typed models generally; they are not predictions about later models.

Table 6: Verdict for PoL2, function by function. *Req.*: requirements of Table 2 involved. *Evidence*: the coded findings that decide the verdict, with the decisive numbers. *Verdict*: NS, not supported; DC, supported only as a recorded detector with conditions (the design of Section 6); NT, not tested on PoL2’s construct. A cell may combine NS for one part of a function with NT for another; “precautionary” marks a restriction that rests on the absence of a measurement rather than on a measured failure (the rule of Section 7.2). *Certainty*: rating of the deciding findings in Table 3; ^c contested; “not rated” marks results that are not key findings and rest on single studies. All ratings are low or very low.

PoL2 function	Req.	What the evidence shows	Verdict	Certainty	What would be needed
Safety valve: blocking hate language in inputs (§ 5.4.1)	G1, G2, G3, G6	Ranks harmful English content well zero-shot (median AUROC 0.886 at \$0.30 against \$18.96 per pass [16]) and on the HateCheck diagnostic set is within 1.6 macro-F1 points of a frontier LLM at 97% lower cost [181]; but fixed thresholds fail (F1 0.158 at 0.5 against 0.947 fitted [126]), a JEV-specific injection and a universal suffix reached 97–100% attack success [184], one inserted opinion flipped 12.1% of decisions [138], and swapping option definitions changed 32.5–50.5% of answers [15, 31].	DC: automatic <i>hold</i> only on agreement of independently built detectors; final block by a person	low (ranking; steering; option names); low ^c (thresholds)	A three-valued Chinese benchmark; thresholds certified on local held-out data and re-certified over time; red-team success per attack type below a target set in advance; neutral identifiers passing a renaming test
Output suppression: hate-language truncation of PAI’s own output (§ 4.3.3.1)	G1, G3, G5	Hate against merely offensive language is separated weakly by every system (three-class macro-F1 JEV 0.532, frontier LLM 0.604, supervised TF-IDF 0.639 [181]); exact selection of the violated hate-speech policy 29.4% without retrieved precedents [180]; errors concentrate in sub-types that aggregates hide (130 of 167 misses at confidence ≥ 0.95 [129]).	DC: automatic <i>repair</i> of the AI’s own output is reversible and within the detector role; the truncation record must show what was removed	low (sub-types; option names); low (ranking)	Measured recall of hate language against offensive language and the state of absence in Chinese; sub-type recall floors; a record of each truncation open to the person addressed
Real-time correction support: identification and correction suggestions (§ 4.3.3.2)	G1, G5	Identification is the row above. A typed decision returns a probability and nothing else [163], so a correction <i>suggestion</i> needs a generative component whose grounds are not those of the verdict (Section 4.5); on Chinese emotion classification every system was weak (macro-F1 JEV 21.06, generative 19.82–23.31 [194]).	NT: the typed interface covers identification only; suggestions untested	low (ranking); not rated (Chinese emotion, single run)	A test of whether suggestions generated after a typed verdict reflect what produced it, in Chinese

Table 6, continued.

PoL2 function	Req.	What the evidence shows	Verdict	Certainty	What would be needed
Anomaly detection and risk warning (Art. 7(2)–(3))	G1, G2, G6	On text, a detector with the conditions above; on non-text signals, zero-shot JEV scored below the majority class on network-flow features (9.8% against 22.8% [132]) and, in industrial fault detection, flagged 57–94% of fault-free windows and under shift did no better than flagging every window [193]; attacks disguised as normal operation had recall 0.34, and 21 of 76 windows an agreement gate accepted as benign were attacks [134].	DC on text; NS on numeric or structured signals without a trained model	low (sub-types; steering); very low (agreement gating)	A definition of “anomalous pattern” in the specification’s own terms; drift and disguised-attack tests; a trained or rule-based detector as the independent second check
Ethical verification and alignment (Art. 6(3))	G3, G4, G5	Without a persona JEV answered value questions like its American personas, and Saudi personas reproduced 87% of the human Saudi–US difference in English but 62% in Arabic [146]; on moral dilemmas rewording moved its answers much more than re-runs did [144] [‡] ; where annotators disagreed most its calibration error rose to 0.339 [156] [‡] , and it reported 0.88 on items on which people split 55 to 45 [206] [‡] .	NS as an automatic verifier, because it depends on a neutral default and on calibration where people disagree, both contradicted; the EAP construct itself NT	very low (culture); low (calibration is local)	A benchmark of value-laden PoL2 questions with disagreement-aware labels, its default position without persona reported, and contested items routed to people
Dispute adjudication without human intervention (Art. 10, Art. 9(7))	G4, G5, G6	Low-cost peer evaluators repeated JEV’s most confident errors in 96.0% of verdicts (an upper bound) [170] and caught at most 5 of 130 confident misses [129]; an explicit “I don’t know” was chosen on 10.5% of the items where it was correct and “None” on 7–54% [151, 160]; the same decision asked through two interfaces led to different actions in 32.8% of cases [158]; a threshold fixed in advance missed its accuracy target on 8 of 14 tasks [19], a finding one preregistered study contests [173]; the output carries no reasons [163].	NS as judge; DC only for ranking and triage of filings	low (peer errors; “I don’t know”); low ^c (thresholds in advance)	The four conditions of “What would change the conclusion” below, met on a PoL2 benchmark in Chinese in one preregistered and independently replicated study, or two independent ones
Per-person quantification of Love Language and hate language (§ 5.6)	G3, G7	Group bias in one informal forced-choice test (“White” 51% of first picks [145] [†]); typed and generative models degraded on low-resource languages [143]; a third party’s inserted opinion flipped 12.1% of decisions [138], so an aggregate could be manipulated; no study measures differential error by speaker group on behavioural judgements.	NS as a decision input (steering, low); NT on error by group: score events and behaviours, never persons, until error by group and language is published (precautionary, Section 7.2)	very low (culture); low (steering); not rated (group bias, single informal test; language degradation, one benchmark)	Published error rates by group and language in Chinese; consent, access to one’s record and appeal; a legal reading against social-scoring prohibitions (Section 5, T5)
Assessment dimension in public decisions (§ 5.6)	G7	Population-scale reading can be done honestly: a shared human–model error term gave retrospective coverage of 0.93 and 0.96 at nominal 80% and 90% [28], and audited probability samples gave corrected statewide counts, but only after per-factor human recalibration [131, 183]; simulated publics lean towards one population [146] and understate disagreement [206] [‡] .	DC: an audited aggregate measurement with its measured error carried into every conclusion; not a decision input on its own	very low (culture); not rated (audit designs, single studies)	A disagreement-aware audit sample of real PoL2 items; the measured gap to people reported with every estimate

The verdict. On the evidence to 8 October 2026, hosted JEV is not a suitable tool for any PoL2 function that decides. That covers dispute adjudication under Art. 10, automatic ethical verification under Art. 6(3), the per-person quantification of § 5.6 (on steering and on the absence of any group-error measurement), and any use of the safety valve in which a block of a person’s speech is final without a person’s confirmation. The decisive findings are negative and, within their low or very low certainty, consistent: thresholds fitted in one setting do not transfer to another (low, contested), decisions can be steered by content inside what the model audits (low; 97–100% attack success for a tailored injection [184], 12.1% of decisions flipped by one inserted opinion [138]), option names can override their definitions (low), an explicit “I don’t know” option is rarely chosen when it is correct and “none” inconsistently (low), and the low-cost evaluators to which such a model would escalate repeated almost all of its confident errors (low, an upper bound). A judge with these properties would restrict speech and allocate welfare on verdicts that an adversary can move, that depend on how the question was worded, and that no second automated opinion is likely to catch; Art. 9(7), which keeps human supervision from intervening, would remove the one check the evidence says is needed. No typed model tested escapes this verdict, and no low-cost generative evaluator tested beside one has been shown to: where the comparison was made, typed models were not generally worse (low, by judgement), but most failures were never compared, so the verdict is not specific to typed models and, until generative evaluators are tested under the same protocol, applies to automated judges of this cost tier as a precaution (Section 7.2).

For the functions that detect rather than decide, the safety valve of § 5.4.1, the truncation of § 4.3.3.1 and the anomaly detection of Art. 7 on text (not on numeric or structured operational signals without a trained model, Table 6), hosted JEV is usable, but only inside the design of Section 6; the public-decision assessment of § 5.6 is usable only as an audited aggregate measurement with its measured error carried into every conclusion; and the positives that support this are real but conditional. It ranks harmful English content well at a small fraction of the cost of generative evaluators (low), on the HateCheck diagnostic set it came within 1.6 macro-F1 points of a frontier model at 97% lower cost [181], it is often well calibrated on tasks that resemble its training (low), confidence-based deferral worked in-distribution [123, 152], and in one preregistered study accepting its low-risk verdicts only when an independently built rule agreed lost no macro-F1, though disguised attacks passed (very low). Each positive comes with the condition that turns it into a design element: ranking needs a locally fitted and re-certified threshold before it is a decision, calibration holds only where it was measured, deferral holds only in-distribution, and agreement gating needs a second detector of different construction. Even this detector verdict is an extrapolation: it rests on English benchmarks of binary hate speech, whereas PoL2 asks for three-valued judgements of Love Language, hate language and the state of absence in Chinese, and no coded study measures that construct; the only Chinese emotion result found every system weak (macro-F1 about 21 [194]), the only Chinese decision benchmarks concern service routing, scam detection and ship-design engineering rather than hate language [148, 195], and hate against merely offensive language, the distinction that § 1.5 most needs, was separated weakly by every system tested [181]. We therefore state the verdict with its certainty attached: every finding above is low or very low, four are contested, most rest on preprints by single teams in the model’s first weeks, and the ratings could change with a single well-designed study on the specification’s own construct.

Two further findings bear on PoL2 beyond any single function. A typed safeguard can also serve as a queryable oracle: with conditional instructions planted in the input, hosted JEV’s labels alone revealed hidden gender, age-group, heart-disease and ethnicity attributes in every tested case, with two to six queries per state (an open model’s labels revealed none); separately, its verdicts raised the judged harmfulness of harmful plans, and its labels trained a surrogate to 92% agreement from 200 labels [184]. This argues for keeping the valve’s probabilities from untrusted callers and keeping

sensitive fields a decision does not need out of its input, which for hosted JEV is the only one of these measures that would remove the signal, since the attack worked for a user authorised for the task and needed labels alone; restricting untrusted callers’ queries to authorised tasks keeps the valve from serving as a general oracle for questions such as the feasibility of harmful plans, and logging makes surrogate extraction through the authorised task visible, though neither prevents it (Section 6, element 4). None of these measures has been tested. And a closed hosted model cannot satisfy Art. 4 and Art. 5(4) as written (T3): its decision records can be published, its reasoning cannot, and the trade against the measured robustness of open models, which showed the most severe name inversion [15], has to be made explicitly.

What transfers. Several lessons hold for PoL2 whatever model is used, because each follows from a finding that concerns the interface or the task rather than JEV.

- *Ask three-valued questions, and ask sufficiency separately.* A yes/no question discards the state of absence (§ 4.3.2); explicit “I don’t know” and “none” options were chosen inconsistently (low), but a separate yes/no question on whether the evidence suffices detected missing information better than confidence in one study [25] (very low). The three-valued verdict is already in the workstream’s pilot protocol [54] (Section 6, element 3; T2).
- *Bind neutral identifiers to definitions and test by renaming.* Option names overrode definitions for every typed model tested (low), and digit or random-string identifiers brought hosted JEV near its noise floor [31]; moral vocabulary, which § 1.5 says its concepts are not, belongs in definitions, not in option names (element 3; checklist item 4).
- *Record every decision.* A typed verdict carries no reasons (Section 4.5), so the record (question, option set and definitions, input hash, probabilities, model and calibration versions, ground cited, reviewer outcome) is the only thing that can be audited or questioned under Art. 9, and the only basis for the verifiable chain of Art. 5(5) (element 7).
- *Calibrate locally and re-validate over time.* Thresholds fitted elsewhere did not transfer (low, contested) and calibration was local (low); hate-speech rankings on static benchmarks did not predict rankings on time-stamped or neologism-substituted data [135], so certification has a date and a language (element 4; checklist items 2, 3, 11 and 16).
- *Gate automatic action on agreement between differently built detectors.* Peer models repeated confident errors (low); accepting a verdict only when an independently built rule agreed lost no macro-F1, though disguised attacks still passed (very low), so independence must be built, not assumed, and its miss rate measured (element 3; checklist item 10).
- *End every chain that could restrict a person with a person, and audit confident decisions.* The most damaging errors were confident and shared [129, 170], and no study compares typed models with trained reviewers on governance items; until one does, a human end-point with a blind random audit is the only check the evidence does not undermine (elements 5 and 6; T4, option (b)).
- *Keep probabilities from untrusted callers, keep unneeded sensitive fields out of the input, restrict and log untrusted queries, and separate independent flags into separate calls.* Probability feedback helped attackers [137], labels alone leaked attributes and trained surrogates [184], and one planted note changed several answers in one call [141][‡] (elements 3 and 4).
- *Score events and behaviours, never persons, until error by group and language is published.* This follows from § 4.3.1 as much as from the evidence (T5).

- *Build the construct benchmark first.* Every verdict above is an extrapolation from English binary tasks; an independently annotated, disagreement-aware, time-stamped benchmark of Love Language, hate language and the state of absence in Chinese, with a held-out set, is the prerequisite for turning any of these verdicts from extrapolation into measurement (open problems below).

7.2 Scope, certainty, limitations and open problems

Beyond one specification. Any proposal that puts an automated filter in front of AI systems, gives disputes to AI, or scores people’s behaviour for public purposes faces the questions G1–G7, and, with the caveats on certainty below, the design conclusions of Section 6 are likely to apply to it. The comparison in Section 2.3 makes this concrete: constitutional classifiers, the Chinese duty to stop unlawful generation and the DSA’s provisions on automated means each move toward an inference-time interception layer, and each then needs an answer to G1–G5 and, where it automates oversight, to the escalation question of G6, which its own text gives at most in part (the Chinese Measures require records of measures taken against users and a complaint channel, Art. 14(2) and 15 [11]); G7 and the rest of G6 arise only where a proposal also gives disputes to AI or scores people. What the case adds is a demonstration that a governance text can be read clause by clause as a set of testable requirements, and that doing so exposes tensions that are invisible while the text is read only as a statement of values. In this case several of the framework’s own commitments, the state of absence, the warning that love and hate are not moral labels, and the distinction between persons and behaviours, are consistent with what the evidence recommends, and the tensions arise where other provisions pull against them.

Certainty. Table 3 rates each finding used in the abstract and conclusion. The rules count every coded study that bears on a finding, count studies sharing an author once, accept as independent replication only another team’s study on its own data or a reimplementation, count evidence per system arm and rate findings about hosted JEV or typed models generally on hosted-model evidence alone, with open-model studies as corroboration, and lower a rating when a study of at least equal design strength points the other way (Section 3.5); every coded study’s assignment to each finding is released (`data/certainty_evidence.csv`), and the ratings are computed from it by `scripts/rate_certainty.py`. These rules were revised during review rounds 12–15, after the update studies had been coded, in response to reviewer findings, and all findings were re-rated with them; this departs from the update protocol, which had frozen the main-window ratings, so the table shows the rating as given on 4 October beside the current one. No finding reaches moderate certainty. That inserted content steers decisions would be moderate if open-model studies could raise a rating; under our rule they only corroborate. Four are contested: that injection rarely selects the attacker’s target, which is very low; that escalation to a stronger reasoner saves cost without loss of accuracy, which is low; and the two threshold findings, that fixed thresholds do not transfer and that thresholds set in advance miss their error targets on new data, which are low. Both are supported by a preregistered study in which a threshold fixed in advance missed its accuracy target on 8 of 14 tasks [19] and by several other studies of JEV, and both are opposed by a preregistered study of changed question types in which JEV’s gate, set in advance, stayed within its target on 6 of 7 templates [173], and the second also by two update studies whose thresholds met their targets [178, 190][‡]; because the preregistered opposing study ranks as high as the strongest supporting one, both fall from moderate to low. All ratings are low or very low, because the sources are preprints posted within days of a model’s release, many results rest on one study, the one reproduction of a headline study re-ran its authors’ code and data, and no study measures the

specification’s own construct.

Limitations. The corpus covers seventeen days of submissions and consists of unrefereed preprints and grey literature, many from single authors working quickly; of the 27 early studies graded by a community tracker, none was rated at low risk of bias [119]. Studies of failures are quick to write in a model’s first weeks and may be over-represented; Zenodo records were negative more often than arXiv papers, although dropping them changes the overall shares little (Section 4.8). Several studies use open models rather than the hosted one, and some do not report the hosted version; we say so each time, but the tallies pool hosted and open models. The requirements are derived from PoL2, so the analysis answers its questions and may miss strengths of typed models that it does not ask about. Screening, extraction, coding and verification were carried out by AI agents under the author’s direction, and both coders were AI agents of the same model family, so the agreement statistic ($\kappa = 0.72$, 95% CI 0.42–0.94, on a sample dominated by Q codes) is an upper bound on how reproducible the codebook is for human coders; the census of all update pairs gave $\kappa = 0.55$ (95% CI 0.36–0.70), with an overlapping interval, and is the better estimate for AI coders, which implies that a share of the singly coded main-window codes would be resolved differently on a second reading. The main tallies cover records retrieved by 4 October, and the main search’s recall was incomplete: 18 of the 103 coded studies dated within the window were found only by the update search of 8 October (Section 4.9), which is reported separately, not pooled, and includes 13 Zenodo records whose earlier exclusion or omission cannot be established, and papers announced after 8 October are not included. No study compares a typed model with trained human reviewers on a governance task, so we cannot say whether its errors are worse than those of the people it would assist.

What would change the conclusion. We would revise “not a judge” if, on a held-out, independently annotated three-valued PoL2 benchmark in Chinese, a typed model with risk-controlled thresholds (i) kept its false-block and miss rates within targets set in advance by those accountable for the deployment, (ii) changed fewer answers under name and definition swaps than twice its repeat-noise floor, (iii) flipped fewer than 5% of decisions under single-opinion insertion and under a 64-query adaptive attack, and (iv) agreed with an independent panel of trained reviewers at least as well as the reviewers agree with each other, in one preregistered study with an independent replication, or in two independent studies. We would revise the “detector” claim if, on such a benchmark, its AUROC fell below 0.75, if two independently built detectors repeated more than 80% of each other’s confident errors, or if the review load at certified thresholds exceeded the reviewer capacity budgeted for it. We ask for more evidence to relax a safeguard than to add one, because a wrongly removed safeguard restricts people’s speech and resources while a wrongly added one costs review time. We would revise the recommendation that consequential decisions end with a person if, on the same benchmark, trained reviewers agreed with the independent panel less well than the gated detector ensemble, or moved their verdicts towards the model’s displayed verdict in more than a pre-set share of audited cases. And if generative evaluators tested under the same protocol passed where typed models failed, the conclusion would become specific to typed models; until then it is not.

Open problems.

- *A PoL2 benchmark.* No public dataset labels Love Language, hate language and the state of absence. An independently annotated benchmark in Chinese, built with disagreement-aware

labels [66] on the model of existing Chinese offensive-language resources [69, 70], with a held-out set kept outside any repository, is the first prerequisite for every recommendation above; the workstream’s 20 public pilot items are explicitly not a ranking set [54]. Because static hate-speech benchmarks lose predictive value as language changes [135], such a benchmark should be time-stamped and periodically refreshed, including paired items with newly emerged terms.

- *Human reviewers versus typed detectors* on the same governance items, including reviewers’ agreement with each other and their susceptibility to the model’s verdict.
- *Comparators for failures*. Most negative findings had no generative comparator; re-running them with one would show which failures are specific to typed models.
- *Robustness of a queryable safeguard*. Whether hiding or coarsening probabilities, adding noise, rate-limiting, restricting queries to authorised tasks or keeping sensitive fields out of the input reduces attack success, and at what cost to legitimate users, has not been measured.
- *Faithful explanations*. Whether an explanation written afterwards by another model reflects what produced a typed verdict needs its own evaluation before it can satisfy Art. 9(2).
- *Per-person measurement*. If § 5.6 is implemented, differential error across groups and languages must be measured before any per-person measure is computed or stored.
- *Replication*. Most findings rest on one study by one team; independent replication on new data (the reproduction of *Just Ask Jev* re-ran the authors’ code and data [126]), is the most valuable next contribution for almost every requirement.

8 Conclusion

Our verdict for PoL2 is explicit and, on current evidence, negative for every function that decides (Section 7.1). Hosted JEV and the typed models tested beside it, and on precaution the low-cost generative evaluators tested beside them (not shown to escape these failures, most of which were never compared), should not adjudicate disputes under Art. 10, verify ethical alignment automatically under Art. 6(3), quantify persons under § 5.6, or make a block of a person’s speech final, because thresholds do not transfer, decisions can be steered by the content they audit and can follow option names rather than definitions, an explicit “I don’t know” option is rarely chosen when correct and “none” inconsistently, and the peer evaluators to which such a model would escalate repeated almost all of its confident errors; for per-person quantification the verdict also rests on precaution, since no study measures error by speaker group. For the functions that detect, the safety valve of § 5.4.1, the truncation of § 4.3.3.1 and the anomaly detection of Art. 7 on text (not on numeric or structured signals without a trained model), hosted JEV is usable only as a recorded, three-valued, locally calibrated *detector* inside the design of Section 6, in which automatic action requires agreement between independently built detectors, any restriction of a person is brief and reversible, every chain that could block ends with a person, confident automatic decisions are audited, and every decision is recorded; the public-decision assessment of § 5.6 is usable in the same sense, as an audited aggregate measurement with its measured error carried into every conclusion. Even that detector verdict is an extrapolation from English benchmarks of binary hate speech: no study measures the specification’s three-valued construct in Chinese, the only Chinese emotion result found every system weak, and hate against merely offensive language was separated weakly by every system tested. The certainty of every finding behind these verdicts is low or very low, four findings are contested, and an update

search to 8 October reversed none of them; a single well-designed study on the specification’s own construct could change any of them.

Typed decision models nonetheless arrived with the interface that automated governance safeguards need: a typed verdict with a probability, for every input and output, at a cost that makes universal screening affordable. What transfers from this assessment to PoL2 regardless of model is a set of practices tied to that interface: three-valued questions with sufficiency asked separately, neutral identifiers bound to definitions and tested by renaming, a record of every decision, local calibration with temporal re-validation, agreement gating between differently built detectors, human-ended chains with blind audit of confident decisions, probabilities kept from untrusted callers, unneeded sensitive fields kept out of the input and untrusted queries restricted and logged, independent flags asked in separate calls, scoring of events rather than persons, and, before any of these can be measured rather than extrapolated, a construct benchmark in Chinese.

For PoL2, several of the framework’s own commitments are consistent with what the evidence recommends: the state of absence, the warning that love and hate are not moral labels, and the distinction between persons and behaviours. Other provisions are sharpened by the evidence into choices the text has yet to make: whether to draw an explicit line between understanding emotion in language and inferring persons’ emotional states; how to ask three-valued questions of models that say “insufficient” unreliably; how to reconcile transparency with closed models; whether non-intervention should yield, for consequential decisions, to a human who can suspend them, close to what data-protection law requires where it applies; and whether per-person assessment can be squared with equal connection. Making these choices explicit, and testing the resulting safeguard on the specification’s own construct, is the work that follows.

Positionality and competing interests

The author is one of the contributors listed in the PoL2 text and contributes to the NaturalDAO governance workstream whose pilot protocol this analysis discusses; the author has a reputational interest in that work and in the framework. Several recommendations in Section 6 coincide with choices already made in that pilot protocol, and each such coincidence is flagged so that readers can judge whether the evidence or the prior design drove it. To limit the effect of this position, the analysis uses a codebook that is symmetric between supporting and negative findings, appraises every coded study, reports certainty for each headline finding, offers at least one text-preserving option for every tension, lists the corrections to its earlier dataset release, and releases all coding with rationales. The tension analysis was not reviewed by the framework’s originator or by other contributors before release. The author has no financial relationship with TypeSafe AI or with the developers of any model discussed.

Use of AI assistance

AI agents (Claude Opus 5.5, Anthropic) carried out the searches, screening, full-text extraction, coding, quality appraisal, second coding, figure scripting and drafting, and checked the text against its sources in separate review passes by fresh agents of the same model; the agents’ instructions, findings and resolutions are released with the data. The author directed the work, set the research question, requirements and codebook, and is responsible for all content; the author did not re-code or re-check individual codes by hand before this release.

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Data and code availability

The supplementary material (Supplements S1–S5: clause concordance, evidence tables, catalogue of studies, corrections and certainty ratings, and codebook, search and selection) is a separate PDF at <https://github.com/naturaldao/NaturalDAO/blob/main/PoL-Governance/research/PoL2-Jev-Typed-Literature-Survey/paper/supplement.pdf> and an ancillary file of the arXiv submission. The catalogue (`data/papers.csv`, `data/zenodo_preprints.csv`, `data/grey_literature.csv`), the evidence coding (`data/evidence_coding.csv`), the quality appraisal (`data/quality_appraisal.csv`), the coder-agreement data (`data/coder_agreement.csv`), the excluded records (`data/excluded_records.csv`), the update-search data of 8 October (`data/papers_update_2026-10-08.csv`, `data/evidence_coding_update_2026-10-08.csv`, `data/quality_appraisal_update_2026-10-08.csv`, `data/coder_agreement_update_2026-10-08.csv`, `data/excluded_records_update_2026-10-08.csv`, `data/search_rerun_2026-10-08.csv`, `data/search_rerun_zenodo_2026-10-08.csv`, `data/search_rerun_2026-10-08.csv`), the certainty evidence (`data/certainty_evidence.csv`, every coded study’s assignment to each key finding per system arm) and the script that computes and checks the certainty ratings (`scripts/rate_certainty.py`), the search scripts, the review and audit logs, the bibliography verification log, the figure scripts and the LaTeX sources are released under CC0 1.0 in the `PoL-Governance/research/PoL2-Jev-Typed-Literature-Survey` directory of the NaturalDAO repository (<https://github.com/naturaldao/NaturalDAO>). Full texts are redistributed only for papers under Creative Commons licences, with their original licences; all others are linked.

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